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Eddy viscosity from stochastic homogenization: exact cell solutions, the two-thirds law from the filtered energy budget, the spectral eddy viscosity of a shock, and tests against direct numerical simulation

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ABSTRACT

We build an eddy viscosity from two-scale stochastic homogenization and test it against exact solutions and direct numerical simulations (DNS). For a top-hat filter of width R the stationary filtered energy budget is exactly $\nu[1+\nu_e(R)]\Omega_R=\varepsilon$, with $\nu\nu_e$ defined by the subfilter flux; in one dimension $\Omega_R=S_2(R)/R^2$. The cell eddy viscosity follows from the advection of resolved momentum by the subcell velocity, a closure that neglects pressure and back-reaction in the cell. In one dimension it is exactly $\nu_e=\int(g-1)\ln g\,\mathrm{d}y\geqslant 0$ with $g=e^{\xi}/Z$: quadratic in the cell Reynolds number when diffusion dominates, $\xi_{\max}-\int\xi-1/\alpha$ when advection dominates. It reproduces the inertial eddy viscosity of a Burgers shock, $u_0R/(6\nu)$, which is also the long-wave spectral eddy viscosity of the shock for a top-hat filter; we extend the inviscid sharp-cutoff result for the shock to finite viscosity and any filter and trace its cusp to nonlocal triads. In the inertial range $\nu_t=cRS_2^{1/2}$, and the budget gives $S_2\propto(\varepsilon R)^{2/3}$; random-phase cells give $c=0.0817$ and, in a separable three-dimensional estimate, $C_K\approx 1.36$. A priori tests on Burgers DNS and on isotropic turbulence at $R_\lambda\approx 420$ confirm the budget to about 10% and the shock predictions to within 30%. Cells built from DNS velocities reproduce the Gaussian coefficient, but the measured coefficient is $c=0.060$; with it the budget gives $C_K\approx 1.67$. The closure misses gradient transfer in the dissipative range and backscatter.

Key words: Turbulence; Kolmogorov law; Eddy viscosity; Homogenization; Burgers equation; Large-eddy simulation

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1 INTRODUCTION

The second-order structure function of the velocity, $S_2(r)=\langle[u(x+r)-u(x)]^2\rangle$, is the simplest measure of how kinetic energy is shared among scales in turbulence. Kolmogorov's 1941 theory (K41) predicts that in the inertial range, far from the forcing scale L and far above the dissipation scale $\eta=(\nu^3/\varepsilon)^{1/4}$, the structure function depends only on the mean dissipation rate ε and the separation r [1, 2]. Dimensional analysis then fixes $S_2=C_2(\varepsilon r)^{2/3}$, or $E(k)=C_K\varepsilon^{2/3}k^{-5/3}$ for the energy spectrum [3, 4]. Measurements and simulations agree with this law up to small intermittency corrections and give $C_K\approx 1.5$–1.6 [5, 6, 7].

Two kinds of argument support K41. The first is exact but incomplete: the Kármán–Howarth equation gives the four-fifths law $S_3=-\frac{4}{5}\varepsilon r$ [2, 8], but it contains two unknowns, S_2 and S_3, and does not determine S_2. The second is complete but not exact: closures replace the unknown energy transfer by a model. Heisenberg's eddy viscosity [9], the local transfer models of Kovasznay and Pao [10, 11], the direct-interaction and EDQNM family [12, 13], renormalization-group methods [14, 15] and random-walk models of the cascade [16] all reproduce $k^{-5/3}$. Lagrangian closures of the direct-interaction family predict the constant without adjustable parameters, $C_K\approx 1.72$ in the Lagrangian renormalized approximation [17, 18]. Most of these approaches need a phenomenological input (a mixing length, a damping time or a dimensional ansatz) that carries much of the result. Obtaining S_2 from the dynamics with every assumption stated, every constant computed and every step tested against data remains open [3, 19].

The central unknown in all of these approaches is the eddy viscosity: the rate at which motions smaller than a scale R drain energy from motions larger than R, per unit of resolved strain. It is the basis of the Smagorinsky model of large-eddy simulation (LES) [20, 21] and, in spectral form, of Kraichnan's spectral eddy viscosity $\nu_t(k|K)$,

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the drain of a resolved mode k by modes above a cutoff K [22]. Closure theories predict that $\nu_t(k|K)$ has a plateau at $k \ll K$ and a cusp as $k \to K$ [22, 23]; simulations confirm the cusp for sharp cutoffs and a plateau for graded filters [24, 25]. A mathematically rigorous line of work computes effective (eddy) diffusivities by homogenization: a field transported by a rapidly varying velocity obeys, on large scales, a diffusion equation whose coefficient follows from a cell problem [26, 27, 28, 29, 30, 31]. Applied to the Navier–Stokes equations themselves, multiscale expansions give the eddy viscosity of a prescribed periodic or random small-scale flow, which can be negative [32, 33]. Homogenization gives effective coefficients for a prescribed small-scale flow; it does not, by itself, say what that flow is.

This paper joins these lines and tests the result. (i) For a top-hat filter of width R the energy budget of the filtered Navier–Stokes equations is exact once the eddy viscosity is defined by the subfilter energy flux; in one dimension it ties $\nu_e(R)$ to $S_2(R)$ without approximation (Section 2). (ii) The eddy viscosity of a cell of size R is computed by two-scale homogenization of the resolved momentum advected by the subcell velocity (Section 3); the cell problem is solved exactly in one dimension for deterministic and stochastic subcell velocities (Sections 4 and 5). (iii) The Burgers shock, a nonlinear solution for which every quantity is known, tests the closure in real space (Section 6) and gives the spectral eddy viscosity $\nu_t(k|K)$ in closed form (Section 7); the long-wave limit of its top-hat spectral eddy viscosity coincides with the real-space result. (iv) Combining (i) and (ii) gives $S_2(R)$ at all scales; in the inertial range this yields the two-thirds law and its constant (Section 8), and a separable three-dimensional estimate gives C_K and the Smagorinsky constant (Section 9). Section 10 links phase coherence in the cell to intermittency. (v) Section 11 tests the predictions a priori against DNS of decaying Burgers turbulence and of forced isotropic turbulence from the Johns Hopkins Turbulence Databases (JHTDB) [34, 35]. Section 12 states what is derived, what is assumed, and the strengths and weaknesses that the tests reveal.

Our main results are:

(i) an exact solution of the one-dimensional cell problem, $\nu_e = \int_0^1 (g-1)\ln g\,\mathrm{d}y \geqslant 0$ with $g = e^{\xi}/Z$, its two universal regimes, the offset rule $-1/\alpha$, a closed form for the mean eddy viscosity of Brownian cells, and a non-monotonic dependence on the correlation length (a "stochastic echo");

(ii) for the Burgers shock: agreement of the cell eddy viscosity with the exact real-space value $u_0R/(6\nu)$; the sharp-cutoff spectral eddy viscosity at finite viscosity, extending the inviscid result of Oberai *et al.* [36]; the nonlocal-triad origin of its logarithmic cusp; a long-wave formula for any filter, which gives $u_0\Delta/6$ for a top hat of width Δ; and the real-space kernel of the sharp-cutoff eddy viscosity;

(iii) a derivation of $S_2 = C_2(\varepsilon R)^{2/3}$ in which the exponent follows from scale invariance and the exact budget, and the constant from a computed cell coefficient $c = 0.0817 \pm 0.0004$, with the three-dimensional estimates $C_K \approx 1.36$ and $C_S \approx 0.190$;

(iv) a link between phase coherence inside a cell and the intermittency correction of the β-model [37];

(v) a priori DNS tests: the three-dimensional budget closes to about 10%; the shock predictions hold in Burgers DNS to within 30% without adjustable parameters; one-dimensional cells built from DNS velocities reproduce the Gaussian coefficient; and the separable closure overestimates the measured three-dimensional eddy viscosity by 10–35% in the inertial range, the measured coefficient being $c = 0.060$.

Table 1. Main symbols.

Symbol	Meaning
ν, ε	kinematic viscosity; mean dissipation rate per unit mass
η, u_η	Kolmogorov length $(\nu^3/\varepsilon)^{1/4}$ and velocity $(\nu\varepsilon)^{1/4}$
R	top-hat filter width = cell size
Δ	width of a general filter
Ω_R	$\langle\lvert\nabla\bar{u}_R\rvert^2\rangle$, resolved gradient variance
Π_R	subfilter energy flux
S_2, S_{2L}	second-order (longitudinal) structure function
ν_t, ν_e	eddy viscosity, $\nu_e = \nu_t/\nu$
y	cell coordinate $x/R \in [0,1]$
$v(y)$	subcell velocity minus its cell mean, in units of ν/R
$\xi(y)$, Z	$\int_0^y v\,\mathrm{d}y'$; $\int_0^1 e^{\xi}\,\mathrm{d}y$
s, σ	cell Reynolds numbers (scale of ξ; rms velocity $\times R/\nu$)
μ	correlation length of the subcell velocity divided by R
Re_R	$RS_2(R)^{1/2}/\nu$ (S_{2L} in three dimensions)
c	inertial cell coefficient, $\nu_t = cRS_2(R)^{1/2}$
u_0	half-jump of a Burgers shock
k, K, ζ	resolved wavenumber, cutoff, $\zeta = k/K$

2 AN EXACT ENERGY BUDGET WITH A TOP-HAT FILTER

Consider statistically homogeneous, stationary turbulence forced at scales much larger than R. Apply a top-hat (box) filter of width R, $\bar{u}_R(\mathbf{x}) = R^{-d}\int_{\mathrm{cube}} u(\mathbf{x}+\mathbf{r})\,\mathrm{d}\mathbf{r}$, to the Navier–Stokes equations,

$$\partial_t\bar{u}_i + \partial_j(\bar{u}_i\bar{u}_j) = -\partial_i\bar{p} + \nu\nabla^2\bar{u}_i - \partial_j\tau_{ij} + \bar{f}_i, \qquad \tau_{ij} = \overline{u_iu_j} - \bar{u}_i\bar{u}_j, \tag{1}$$

where τ_{ij} is the subfilter stress. Multiplying by $\bar{u}_i$ and averaging, the stationary resolved energy budget is

$$\langle\bar{f}_i\bar{u}_i\rangle = \nu\left\langle\lvert\nabla\bar{u}_R\rvert^2\right\rangle + \Pi_R, \qquad \Pi_R = -\langle\tau_{ij}\bar{S}_{ij}\rangle, \tag{2}$$

where $\bar{S}_{ij}$ is the resolved strain rate and Π_R is the energy flux from scales larger than R to scales smaller than R. When the forcing acts only at scales much larger than R, $\langle\bar{f}_i\bar{u}_i\rangle = \langle f_iu_i\rangle = \varepsilon$. We define the eddy viscosity $\nu_t(R) = \nu\,\nu_e(R)$ as the flux per unit of resolved gradient variance,

$$\Pi_R \equiv \nu_t(R)\,\Omega_R, \qquad \Omega_R \equiv \left\langle\lvert\nabla\bar{u}_R\rvert^2\right\rangle = 2\langle\bar{S}_{ij}\bar{S}_{ij}\rangle. \tag{3}$$

This is a definition, not a model. With it, Eq. (2) becomes the exact relation

$$\nu\,[1+\nu_e(R)]\,\Omega_R = \varepsilon. \tag{4}$$

The energy injected at large scales is either dissipated by the resolved motions ($\nu\Omega_R$) or passed to the subfilter motions ($\nu_t\Omega_R$). For $R \to 0$, $\nu_e \to 0$ and Eq. (4) reduces to $\nu\left\langle\lvert\nabla u\rvert^2\right\rangle = \varepsilon$. Section 11.3 checks Eq. (4) in DNS.

2.1 The resolved gradient in terms of S_2

In one dimension the derivative of a box-filtered field is a difference quotient,

$$\frac{\partial\bar{u}_R}{\partial x} = \frac{u(x+R/2)-u(x-R/2)}{R} \quad\Rightarrow\quad \Omega_R = \frac{S_2(R)}{R^2} \tag{5}$$

exactly. The one-dimensional budget is therefore a closed relation between the eddy viscosity and the structure function,

$$[1+\nu_e(R)]\,\frac{S_2(R)}{R^2} = \frac{\varepsilon}{\nu}. \tag{6}$$

For $\nu_e \ll 1$ (small R), $S_2 = \varepsilon R^2/\nu$, the exact dissipative-range result. For $\nu_e \gg 1$, $\nu_t S_2(R)/R^2 = \varepsilon$: the flux through scale R equals the dissipation. If ν_t were known as a function of S_2, Eq. (6) would determine $S_2(R)$ at every scale.

In d dimensions, with $\hat{G}(\mathbf{k}) = \prod_i \mathrm{sinc}(k_i R/2)$ the transfer function of the cube filter and $E(k)$ the energy spectrum normalized by $\frac{1}{2}\langle|u|^2\rangle = \int_0^\infty E\,\mathrm{d}k$, isotropy gives

$$\Omega_R = 2\int_0^\infty E(k)\,k^2 \left\langle \hat{G}^2 \right\rangle_{\text{angle}} (kR)\,\mathrm{d}k, \tag{7}$$

where $\langle\cdot\rangle_{\text{angle}}$ is the average over directions of $\mathbf{k}$. For $E = C_K\varepsilon^{2/3}k^{-5/3}$ this is $\Omega_R = 2C_K I_{\text{cube}}\varepsilon^{2/3}R^{-4/3}$ with $I_{\text{cube}} = \int_0^\infty x^{1/3}\left\langle \hat{G}^2\right\rangle_{\text{angle}}(x)\,\mathrm{d}x = 3.373$ (for a ball of diameter R, $I_{\text{ball}} = 4.650$; the equivalent sharp spectral cutoff is $KR = 3.93$). For the ball, $\Omega_R = 3S_2^{\times}(R)/R^2$ with $S_2^{\times} = 24\int_0^\infty E\,j_1^2(kR/2)\,\mathrm{d}k$, and the isotropic relations between the structure functions give, for $S_{2L} \propto R^a$, $\Omega_R R^2/S_{2L} = 3[12a(a-1)+48a]/[a(a-1)+7a+8]$. This equals 15 for $a = 2$, the isotropic dissipative value $\langle|\nabla u|^2\rangle = 15\langle(\partial_1 u_1)^2\rangle$, and $99/14 = 7.071$ for $a = 2/3$, in agreement with $2I_{\text{ball}}/(C_{2L}/C_K)$ computed by quadrature.

3 EDDY VISCOSITY FROM TWO-SCALE HOMOGENIZATION

3.1 The cell problem

Inside a cell of size R the resolved velocity w is carried by the subcell velocity. We model this by the advection–diffusion of the resolved momentum by a prescribed subcell velocity $\mathbf{v}$,

$$\partial_t w + \mathbf{v}\cdot\nabla w = \nu\nabla^2 w, \tag{8}$$

where $\mathbf{v}$ is the velocity relative to the cell mean (the cell mean only sweeps the cell and is removed by Galilean invariance), is statistically stationary, and has zero mean over the cell. Equation (8) is the modelling assumption of the closure (assumption A1): it keeps the advection of resolved momentum by the subcell motion and drops the pressure and the back-reaction of w on $\mathbf{v}$ inside the cell. It is the passive-vector reduction of the convection of microstructure studied by McLaughlin *et al.* [27]. A1 plays the role that the mixing-length hypothesis plays in older closures; Section 11.3 measures its error.

Let $\delta = R/L \ll 1$ be the ratio of cell size to the scale of the resolved field, $\mathbf{y} = \mathbf{x}/\delta$ the fast variable and $\mathbf{v} = \mathbf{v}(\mathbf{y})/\delta$ a strong subcell velocity. Expanding $w = w_0(\mathbf{x},t) + \delta w_1(\mathbf{x},\mathbf{y},t) + \delta^2 w_2 + \dots$ with $\nabla = \nabla_{\mathbf{x}} + \delta^{-1}\nabla_{\mathbf{y}}$ gives [26, 31]

$$\delta^{-2}: \quad \nu\nabla_y^2 w_0 - \mathbf{v}\cdot\nabla_y w_0 = 0 \;\Rightarrow\; w_0 = w_0(\mathbf{x},t), \tag{9}$$

$$\delta^{-1}: \quad \nu\nabla_y^2 w_1 - \mathbf{v}\cdot\nabla_y w_1 = \mathbf{v}\cdot\nabla_x w_0 \;\Rightarrow\; w_1 = \eta(\mathbf{y})\cdot\nabla_x w_0, \tag{10}$$

$$\delta^{0}: \quad \partial_t w_0 = \nu\nabla_x^2 w_0 - \langle \mathbf{v}\otimes\eta\rangle_{\text{cell}} : \nabla_x\nabla_x w_0. \tag{11}$$

The vector η solves the cell problem

$$\nu\nabla_y^2\eta - \mathbf{v}\cdot\nabla_y\eta = \mathbf{v} \quad \text{in the cell,} \qquad \eta = 0 \text{ on its boundary,} \tag{12}$$

and the cell average in Eq. (11) is taken after the zero-boundary condition removes the terms $\langle\nabla_y\eta\rangle$ and $\langle\nabla_y\cdot(\mathbf{v}w_2)\rangle$. The resolved field diffuses with the effective viscosity tensor $\nu\delta_{jk} - \langle v_j\eta_k\rangle$; its isotropic part defines the eddy viscosity

$$\nu_t = -\frac{1}{d}\langle\mathbf{v}\cdot\eta\rangle_{\text{cell}}, \qquad \nu_e = \nu_t/\nu. \tag{13}$$

The expansion is in δ only; the cell Reynolds (Péclet) number is arbitrary. If the cell-mean velocity is kept in $\mathbf{v}$, the same expansion produces, in addition, an effective advection velocity of the resolved field (the analogue of Taylor dispersion); subtracting the cell mean is equivalent to working in the frame that moves with the cell.

We use the zero (Dirichlet) condition on the cell boundary (assumption A2): it fixes the resolved field on the cell faces to its local mean gradient, consistent with a top-hat filter whose cells tile the domain. Measuring lengths in units of R, velocities in units of ν/R and η in units of R, the one-dimensional cell problem becomes

$$\eta'' - v\eta' - v = 0, \quad \eta(0) = \eta(1) = 0, \quad \nu_e = -\int_0^1 v\,\eta\,\mathrm{d}y. \tag{14}$$

In one dimension Eq. (14) keeps the advective form of Eq. (8); the subcell velocity is then compressible, as in the Burgers equation [38, 39].

4 THE ONE-DIMENSIONAL CELL PROBLEM: EXACT SOLUTION

4.1 General solution

Equation (14) can be written $(\eta'+1)' = v(\eta'+1)$. With $\xi(y) = \int_0^y v\,\mathrm{d}y'$ (the integrated subcell velocity, a displacement potential),

$$\eta'(y) + 1 = \frac{e^{\xi(y)}}{Z}, \qquad Z = \int_0^1 e^{\xi(y)}\mathrm{d}y, \tag{15}$$

where $1/Z$ enforces $\eta(1) = 0$. Inserting into $\nu_e = -\int v\eta\,\mathrm{d}y = \int \xi\,\eta'\,\mathrm{d}y$ (the boundary terms vanish) gives the exact eddy viscosity of any one-dimensional cell:

$$\nu_e = \int_0^1 \frac{e^{\xi}}{Z}\,\xi\,\mathrm{d}y - \int_0^1 \xi\,\mathrm{d}y = \int_0^1 (g-1)\ln g\,\mathrm{d}y \;\geqslant 0, \qquad g = \frac{e^{\xi}}{Z}. \tag{16}$$

The second form follows from $\xi = \ln g + \ln Z$ and $\int_0^1 (g-1)\mathrm{d}y = 0$; since $(g-1)\ln g \geqslant 0$ pointwise, the cell eddy viscosity is never negative. If $\xi = sB(y)$ with a fixed profile B and amplitude s, the first term of Eq. (16) is $\partial\ln Z/\partial\ln s$, so

$$\nu_e = s\frac{\mathrm{d}\ln Z}{\mathrm{d}s} - \int_0^1 \xi\,\mathrm{d}y. \tag{17}$$

Z is the partition function of a particle in the random potential $-\xi$; the cell problem is the steady-flux problem of a one-dimensional disordered (Sinai-type) chain [40, 41], and $\ln Z$ plays the role of a free energy. Equations (15)–(16) agree with a direct finite-difference solution of Eq. (14); the largest relative difference over the test profiles below is $2e-06$.

As s grows, e^{ξ}/Z concentrates at the maximum of ξ, so $\eta' \to -1$ everywhere except for a jump of height one at the position $y_{\max}$ of the maximum (Fig. 1). Then $\eta \to -y$ for $y < y_{\max}$ and $\eta \to 1-y$ for $y > y_{\max}$, and Eq. (16) reduces to

$$\nu_e \to \xi_{\max} - \int_0^1 \xi\,\mathrm{d}y - \frac{1}{\alpha}, \tag{18}$$

where the last term is the first correction from Laplace's method: if $\xi_{\max} - \xi \propto |y - y_{\max}|^{\alpha}$ near the maximum, then $\ln Z = \xi_{\max} - \alpha^{-1}\ln s + O(1)$ and Eq. (17) gives $-1/\alpha$. The offset is -1 for a kink ($\alpha = 1$), $-1/2$ for a smooth maximum ($\alpha = 2$) and -2 for a Brownian path ($\alpha = 1/2$). For small s, expanding $e^{\xi}/Z = 1 + \xi - \int\xi + O(\xi^2)$ gives

$$\nu_e \simeq \int_0^1 \xi^2\mathrm{d}y - \left(\int_0^1 \xi\,\mathrm{d}y\right)^2, \tag{19}$$

the spatial variance of ξ over the cell. The cell eddy viscosity is

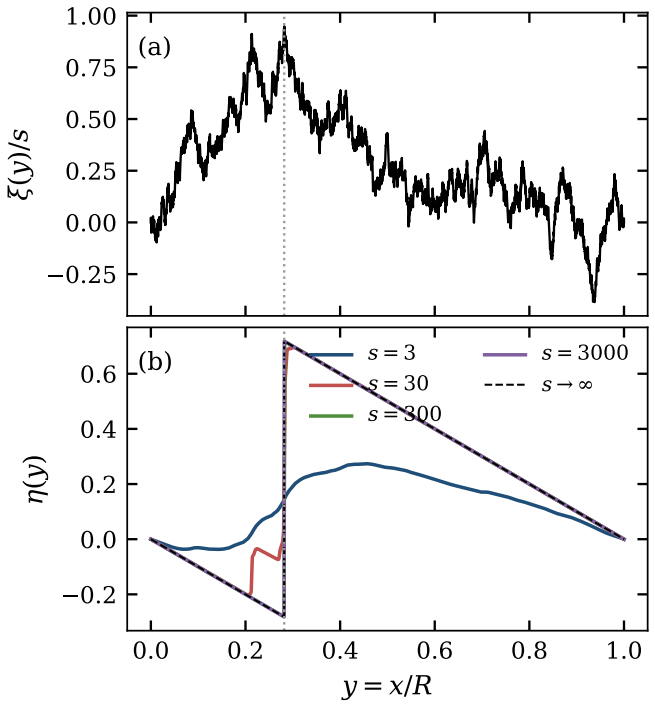


Figure 1. Solution of the one-dimensional cell problem for one realization of a pinned Brownian subcell displacement $\xi = sB$. (a) Profile $B(y) = \xi/s$; the dotted line marks its maximum. (b) Cell solution $\eta(y)$ for $s = 3$ to 3000. As s grows, η approaches the sawtooth $-y$ ($y < y_{\max}$), $1-y$ ($y > y_{\max}$) with a unit jump at the maximum of ξ (dashed), and the eddy viscosity approaches $\xi_{\max} - \int \xi \, \mathrm{d}y$ (Eq. 18).

therefore quadratic in the subcell velocity when diffusion dominates and linear when advection dominates. This quadratic-to-linear transition is the central property used below.

4.2 Deterministic profiles

For a step, $v = -h/2$ for $y < 1/2$ and $v = h/2$ for $y > 1/2$,

$$\nu_e(h) = \frac{h}{4\left(e^{h/4}-1\right)} - 1 + \frac{h}{8}, \tag{20}$$

which is even in h, equals $h^2/192$ for $|h| \ll 1$ and $|h|/8 - 1$ for $|h| \gg 1$ (kink: $\alpha = 1$). For a linear profile $v = h(y - \frac{1}{2})$, diverging for $h > 0$ and converging for $h < 0$,

$$\nu_e(h) = -\frac{h}{24} - \frac{1}{2} + \begin{cases} \sqrt{\dfrac{h}{8\pi}}\,\dfrac{e^{h/8}}{\operatorname{erfi}\sqrt{h/8}}, & h > 0, \\ \sqrt{\dfrac{|h|}{8\pi}}\,\dfrac{e^{-|h|/8}}{\operatorname{erf}\sqrt{|h|/8}}, & h < 0, \end{cases} \tag{21}$$

with $\nu_e \simeq h^2/720$ for $|h| \ll 1$. At large $|h|$ a converging cell has a smooth interior maximum of ξ and $\nu_e \simeq |h|/24 - 1/2$; a diverging cell has its maxima at the two walls, where ξ falls linearly, and $\nu_e \simeq h/12 - 1$. Both offsets follow the rule $-1/\alpha$ (Fig. 2a). For a single Fourier mode $v = \frac{h}{2}\sin[k(y - \frac{1}{2})]$ and small h,

$$\nu_e \simeq \frac{h^2}{4k^2}\left[\frac{1}{2} + \frac{\sin k}{2k} - \frac{\sin^2(k/2)}{(k/2)^2}\right], \tag{22}$$

and for $v = \frac{h}{2}\{\cos[k(y-\frac{1}{2})] - \sin(k/2)/(k/2)\}$, $\nu_e \simeq \frac{h^2}{4k^2}[\frac{4\cos k}{k^2} + \frac{3\sin k}{2k} - \frac{4}{k^2} + \frac{4-\cos k}{6}]$ (both checked against Eq. 16 to 10^{-5}). Averaged over a random phase they give the spectral kernels of Section 5.2.

5 STOCHASTIC CELLS

5.1 Free and pinned Brownian cells

Let the subcell velocity be a random field with zero mean. Its integral starts at $\xi(0) = 0$. If v is measured relative to the cell mean, as Galilean invariance requires, then also $\xi(1) = 0$: the displacement is *pinned*. For a short-correlated velocity of variance σ^2 and correlation length $\mu \ll 1$ (in units of R), ξ is a Brownian bridge, $\xi = sB^\circ(y)$ with $\langle \xi(y)^2 \rangle = s^2 y(1-y)$ and $s^2 = 2\sigma^2\mu$. For comparison we also consider the *free* case, $\xi = sW(y)$ with W a standard Brownian motion, which corresponds to a velocity not measured relative to its cell mean.

For the free case the mean eddy viscosity has a closed form. Using $\langle \ln Z \rangle$ in Eq. (17), the moments of the exponential functional $Z = \int_0^1 e^{sW}\,\mathrm{d}y$ [41, 42] and the Kontorovich–Lebedev transform [43] (Appendix A),

$$\langle \nu_e \rangle_{\text{free}} = 4\int_0^\infty x\, e^{-x^2} \coth\left(\frac{\sqrt{2}\,\pi x}{s}\right)\mathrm{d}x - 2. \tag{23}$$

Its limits are $s^2/6$ for $s \ll 1$ and $s\sqrt{2/\pi} - 2$ for $s \gg 1$: the first is Eq. (19) with $\langle \int W^2 \rangle - \langle (\int W)^2 \rangle = \frac{1}{2} - \frac{1}{3}$, the second Eq. (18) with $\langle W_{\max} \rangle = \sqrt{2/\pi}$ and the Brownian offset -2. Monte Carlo averages over 4000 realizations confirm Eq. (23); at $s = 100$ the simulation gives 77.94 ± 0.42 against 77.81 (Fig. 2b). For the pinned case

$$\langle \nu_e \rangle_{\text{pinned}} \simeq \frac{s^2}{12} \quad (s \ll 1), \qquad \langle \nu_e \rangle_{\text{pinned}} \simeq s\sqrt{\frac{\pi}{8}} - 2 \quad (s \gg 1), \tag{24}$$

because $\langle \int B^{\circ 2} \rangle - \langle (\int B^\circ)^2 \rangle = \frac{1}{6} - \frac{1}{12}$ and the maximum of a Brownian bridge has the Rayleigh law $P(B^\circ_{\max} > m) = e^{-2m^2}$ with mean $\sqrt{\pi/8}$ [44]. At $s = 100$ the simulation gives 61.08, within 1% of 60.67. Pinning lowers the eddy viscosity by a factor two in the diffusive regime and by $\pi/4$ in the advective regime.

5.2 Finite correlation length: the stochastic echo

If $\langle v(y)v(y') \rangle = \int_{-\infty}^{\infty} E_c(k)\cos k(y-y')\,\mathrm{d}k$ (k in units of $1/R$), the small-σ limit of a pinned cell is

$$\langle \nu_e \rangle \simeq \int_{-\infty}^{\infty} E_c(k)\,[X_4(k) - X_3(k)]\,\mathrm{d}k, \tag{25}$$

$$X_4 = \frac{5+\cos k}{3k^2} - \frac{4(1-\cos k)}{k^4}, \qquad X_3 = \frac{(k\cos\frac{k}{2} - 2\sin\frac{k}{2})^2}{k^4}, \tag{26}$$

where X_4 is the contribution of $\langle \int \xi^2 \rangle$ and X_3 that of $\langle (\int \xi)^2 \rangle$. Small wavenumbers ($kR \ll 1$) contribute $\propto k^2$ and large ones $\propto k^{-2}$: only motions of size comparable to the cell transfer momentum efficiently. In the advective limit the same language applies to the variance of ξ at the cell centre, $\sigma_\xi^2 = \int E_c(k) X_1(k)\,\mathrm{d}k$ with $X_1 = 4\sin^4(k/4)/k^2$, and Eq. (18) becomes $\langle \nu_e \rangle \simeq \gamma\sigma_\xi - 1/\alpha$, where $\gamma = \langle \xi_{\max} \rangle / \sigma_\xi$ depends only on the shape of the covariance: $\gamma = \sqrt{\pi/2} = 1.253$ for the Brownian bridge and 0.568 for the self-similar $H = 1/3$ cells of Section 8. Because the large-σ cell solution does not depend on the amplitude of v, the eddy viscosities of velocity components whose displacements share the same maximum add linearly; at small σ they add quadratically.

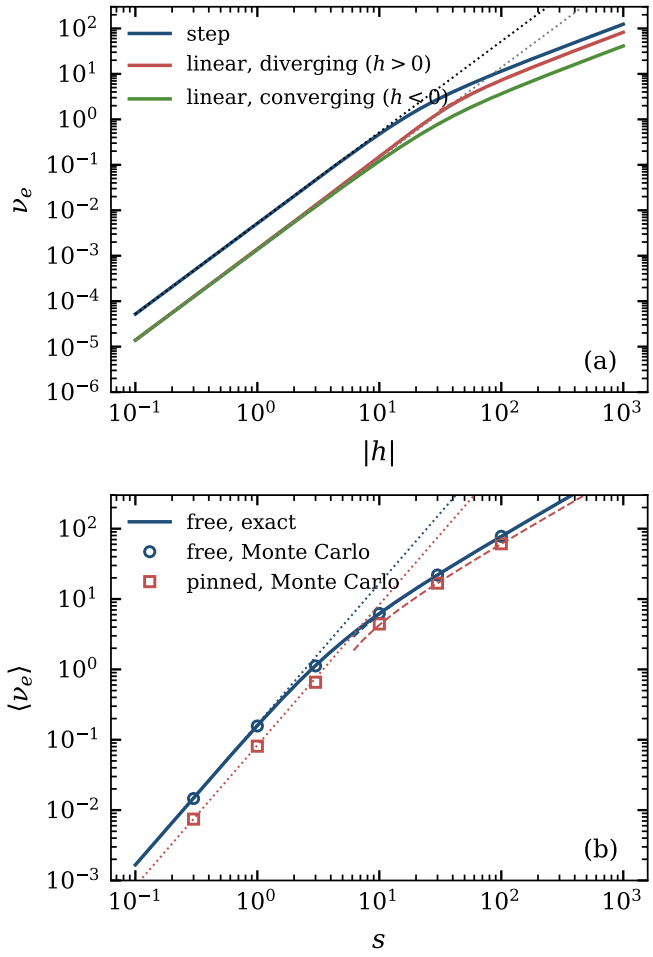


Figure 2. Cell eddy viscosity against cell Reynolds number. (a) Deterministic profiles: step (Eq. 20) and diverging and converging linear profiles (Eq. 21); dotted lines $h^2/192$ and $h^2/720$, dashed lines the large-$|h|$ asymptotes $|h|/8-1$, $h/12-1$ and $|h|/24-1/2$. (b) Stochastic cells: free Brownian displacement, exact result Eq. (23) (line) and Monte Carlo (circles); pinned displacement, Monte Carlo (squares). Dotted: $s^2/6$ and $s^2/12$; dashed: $s\sqrt{2/\pi}-2$ and $s\sqrt{\pi/8}-2$.

For an exponential covariance, $\sigma^2 e^{-|r|/\mu}$, the pinned eddy viscosity rises as $\sigma^2\mu/6$ for $\mu\ll 1$, peaks at $\mu^*\approx 0.18$ with $\langle\nu_e\rangle\approx 0.0093\,\sigma^2$, and falls as $\sigma^2/(180\mu)$ for $\mu\gg 1$ (Fig. 3). The decay at large μ is set by the small-separation behaviour of the correlation: exponential (rough) correlation gives μ^{-1}, Gaussian (smooth) correlation μ^{-2}. Without pinning there is no peak: the eddy viscosity saturates at $\sigma^2/12$. The peak is a "stochastic echo": a quantity that depends only on fluctuations relative to the cell mean vanishes both when the cell contains many independent eddies and when it contains a fraction of one.

6 THE BURGERS SHOCK IN REAL SPACE

The steady viscous shock of the Burgers equation, $\partial_t u + u\partial_x u = \nu\partial_x^2 u$, is $u = -u_0\tanh[u_0 x/(2\nu)]$. Every term of Eq. (6) is known for it, so it tests the closure without statistical assumptions. Integrating over the line instead of averaging, the total dissipation is

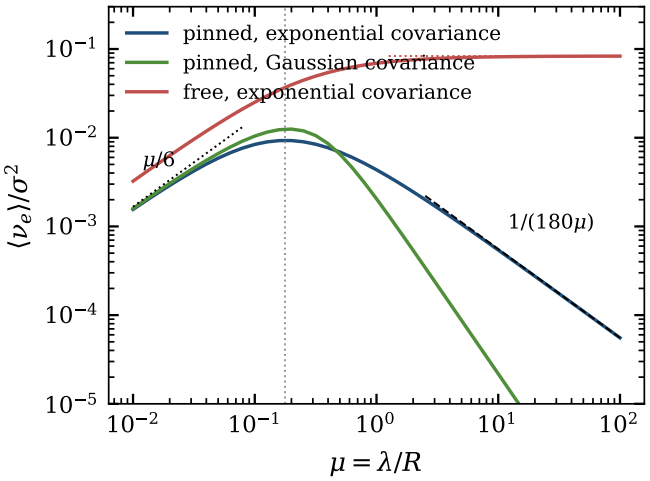


Figure 3. Small-σ cell eddy viscosity $\langle\nu_e\rangle/\sigma^2$ against the correlation ratio $\mu=\lambda/R$. Pinned cells show a maximum near $\mu^*\approx 0.18$ (dotted vertical line) with asymptotes $\mu/6$ and $1/(180\mu)$ for exponential covariance; a Gaussian covariance decays as μ^{-2}. Free cells saturate at $1/12$.

$\nu\int u_x^2\mathrm{d}x = \frac{2}{3}u_0^3$ and

$$\int [u(x+R)-u(x)]^2\,\mathrm{d}x = 4\left[u_0^2 R\coth\left(\frac{u_0 R}{2\nu}\right)-2\nu u_0\right]. \tag{27}$$

The exact budget Eq. (6) defines the exact eddy viscosity of the shock,

$$\hat\nu(R)=\frac{\frac{2}{3}u_0^3R^2}{\nu\int[u(x+R)-u(x)]^2\mathrm{d}x}-1\simeq\begin{cases}\dfrac{1}{60}\left(\dfrac{u_0R}{\nu}\right)^2, & u_0R\ll\nu,\\[2mm] \dfrac{u_0R}{6\nu}-\dfrac{2}{3}, & u_0R\gg\nu.\end{cases} \tag{28}$$

The homogenized prediction is the cell eddy viscosity Eq. (16) evaluated for $v=(R/\nu)[u(x)-\bar u_{\mathrm{cell}}]$ in a cell $[x_0,x_0+R]$, averaged over cell positions with the weight $[u(x_0+R)-u(x_0)]^2$ with which each cell contributes to $\int(\partial_x\bar u)^2\mathrm{d}x$. For $u_0R\gg\nu$ a cell containing the shock at the fractional position a has a step subcell velocity, $v=(2u_0R/\nu)(1-a)$ for $y<a$ and $-(2u_0R/\nu)a$ for $y>a$; ξ rises linearly to $\xi_{\max}=(2u_0R/\nu)a(1-a)$ and falls back to zero. Equation (18) gives $\nu_e=(u_0R/\nu)a(1-a)$, and the average over a uniformly distributed a is

$$\langle\nu_e\rangle=\frac{u_0R}{\nu}\int_0^1 a(1-a)\,\mathrm{d}a=\frac{u_0R}{6\nu}, \tag{29}$$

identical to the leading term of Eq. (28). Figure 4 confirms this over four decades: at $u_0R/\nu=1000$ the computed cell average is $\langle\nu_e\rangle/(u_0R/\nu)=0.1653$ against $1/6$. The constant offsets differ (-1.35 for the cell, -0.67 for the exact budget); they are subleading.

In the dissipative range the two differ in kind. The exact $\hat\nu\propto R^2$ comes from the gradient (Leonard) part of the subfilter stress, $\tau\simeq(R^2/12)u_x^2$, which gives $\Pi_R=-\frac{R^2}{24}\int u_x^3\mathrm{d}x$ and the coefficient $1/60$ [45, 46]. The cell sees a linear subcell velocity, $h=R^2u_x/\nu$, and Eq. (21) gives $\nu_e\simeq h^2/720$; weighting with u_x^2 and using $\int u_x^4\mathrm{d}x/\int u_x^2\mathrm{d}x=\frac{6}{35}u_0^4/\nu^2$ yields $\langle\nu_e\rangle=\frac{1}{4200}(u_0R/\nu)^4$, much smaller. The homogenized closure describes advective transfer by subcell eddies, which dominates the inertial range, and omits the gradient transfer of a smooth field, which matters where $\nu_e\ll 1$. The DNS of Section 11 show the same omission in Burgers and in three-dimensional turbulence.

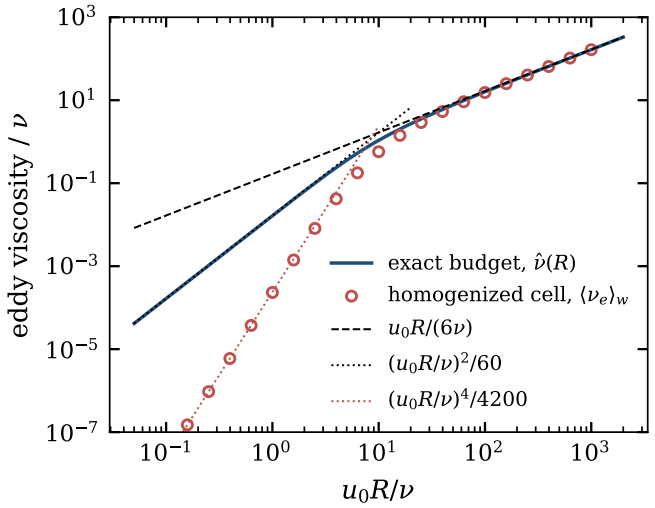


Figure 4. Eddy viscosity of the steady Burgers shock against u_0R/ν. Line: exact budget, Eq. (28). Circles: homogenized cell eddy viscosity, Eq. (16), averaged over cell positions with weight $(\Delta u)^2$. Both approach $u_0R/(6\nu)$ (dashed). In the dissipative regime the exact value follows the gradient-transfer law $(u_0R/\nu)^2/60$ (black dotted) and the cell value $(u_0R/\nu)^4/4200$ (red dotted).

7 THE BURGERS SHOCK IN SPECTRAL SPACE

The real-space test uses a top-hat filter. Large-eddy simulation in spectral space uses the spectral eddy viscosity $\nu_t(k|K)$, the energy transfer into a resolved mode k from triads involving at least one mode beyond the cutoff K, divided by $2k^2E(k)$ [9, 22]. For the shock this quantity can be computed exactly; the calculation shows which triads produce the plateau and the cusp and how both depend on the filter.

7.1 Transfer function of the shock

With $\hat{u}(k)=\frac{1}{2\pi}\int u(x)e^{ikx}\mathrm{d}x$, the shock of Section 6 has

$$\hat{u}(k)=-i\mathscr{A}(k),\qquad \mathscr{A}(k)=\frac{\nu}{\sinh(\pi\nu k/u_0)}, \tag{30}$$

understood as a principal value at $k=0$. Its spectrum, defined by $E(k)=2\pi|\hat{u}(k)|^2$, is

$$E(k)=\frac{2\pi\nu^2}{\sinh^2(\pi\nu k/u_0)}\simeq\begin{cases}2u_0^2/(\pi k^2), & k\ll u_0/\nu,\\ 8\pi\nu^2e^{-2\pi\nu k/u_0}, & k\gg u_0/\nu.\end{cases} \tag{31}$$

The k^{-2} range reflects the jump, the exponential range the shock thickness ν/u_0. The energy equation of mode k is $\partial_tE(k)=T(k)-2\nu k^2E(k)$ with

$$T(k)=\int_{-\infty}^{\infty}T(k,p)\,\mathrm{d}p,\qquad T(k,p)=2\pi k\,\mathscr{A}(k)\,\mathscr{A}(p)\,\mathscr{A}(k-p), \tag{32}$$

the contribution of the triad $(k,p,q=k-p)$, from $\mathscr{F}(u\partial_xu)=-\frac{ik}{2}\int\hat{u}(p)\hat{u}(k-p)\,\mathrm{d}p$. Stationarity, $T(k)=2\nu k^2E(k)$, is the identity $\int\mathscr{A}(p)\mathscr{A}(k-p)\,\mathrm{d}p=2\nu k\,\mathscr{A}(k)$ (principal values at $p=0$ and $p=k$), which follows from

$$\int\frac{\mathrm{d}p}{\sinh p\,\sinh(k-p)}=\frac{1}{\sinh k}\ln\left|\frac{\sinh p}{\sinh(p-k)}\right| \tag{33}$$

and $\lim_{p\to\pm\infty}\ln|\sinh p/\sinh(p-k)|=\pm k$.

7.2 Sharp cutoff: exact result at finite viscosity

For a sharp cutoff the resolved modes are $|k|<K$. For $0<k<K$ the triads with both legs resolved are those with $k-K<p<K$; the others involve at least one subgrid mode and transfer

$$T^{>}(k|K)=\left(\int_{-\infty}^{k-K}+\int_K^{\infty}\right)T(k,p)\,\mathrm{d}p=2\int_K^{\infty}T(k,p)\,\mathrm{d}p, \tag{34}$$

the two tails being equal by the symmetry $p\leftrightarrow k-p$. With $T^{>}(k|K)=-2\nu_t(k|K)k^2E(k)$, $k^*=\pi\nu k/u_0$, $K^*=\pi\nu K/u_0$ and Eq. (33),

$$\frac{\nu_t(k|K)}{\nu}=-1-\frac{1}{k^*}\ln\frac{\sinh(K^*-k^*)}{\sinh K^*}. \tag{35}$$

Equation (35) agrees with principal-value quadrature of Eq. (34) to all printed digits for K^* from 0.16 to 9.4 and k/K up to 0.99. Three properties follow. (i) $\nu_t(k|K)>0$ for all $0<k<K$, because $\sinh(K^*-k^*)<e^{-k^*}\sinh K^*$: a single shock has no backscatter. (ii) At long waves $\nu_t(0|K)/\nu=\coth K^*-1$, which decreases from $u_0/(\pi\nu K)$ for a cutoff in the k^{-2} range to $2e^{-2K^*}$ for a cutoff in the dissipation range. (iii) As $k\to K$ the eddy viscosity diverges logarithmically for any K^*.

When the cutoff lies in the k^{-2} range, $K^*\ll1$, Eq. (35) becomes independent of ν,

$$\nu_t(k|K)=\frac{u_0}{\pi K}\Lambda(\zeta),\qquad \zeta=\frac{k}{K},\qquad \Lambda(\zeta)=-\frac{\ln(1-\zeta)}{\zeta}=1+\frac{\zeta}{2}+\frac{\zeta^2}{3}+\dots, \tag{36}$$

the inviscid result of Oberai *et al.* [36]: a plateau $\nu_t^+=u_0/(\pi K)$ and a logarithmic cusp, $\Lambda(0.5)=1.39$, $\Lambda(0.9)=2.56$, $\Lambda(0.99)=4.65$. Figure 5 shows Eq. (35) for several K^*. For comparison, the normalized three-dimensional EDQNM fit of Chollet and Lesieur [23], $1+34.7\,e^{-3.03K/k}$, has a flatter plateau and reaches a finite value 2.68 at $k=K$. For a train of shocks of mean spacing ℓ, $E(k)\simeq 2u_0^2/(\pi\ell k^2)$ and the plateau takes the Chollet–Lesieur form

$$\nu_t^+=\frac{u_0}{\pi K}=\left(\frac{K\ell}{2\pi}\right)^{1/2}\left[\frac{E(K)}{K}\right]^{1/2}, \tag{37}$$

with a prefactor that depends on the number of shock spacings per cutoff wavelength: the spectral signature of the intermittency of Burgers turbulence (Section 10).

7.3 Origin of the cusp: nonlocal triads

In the inertial limit $\mathscr{A}(p)\simeq u_0/(\pi p)$, and the transfer to mode k from subgrid modes $K<p<P$ is proportional to $\int_K^P\mathrm{d}p/[p(p-k)]=k^{-1}\ln[(P-k)K/(P(K-k))]$. The fraction of $\nu_t(k|K)$ contributed by subgrid modes below P is

$$F(P)=\frac{\ln[(P-k)K/(P(K-k))]}{\ln[K/(K-k)]}. \tag{38}$$

For $k\ll K$, F is spread over $P-K\sim K$. For $k\to K$ the contribution concentrates just above the cutoff: at $k=0.99K$ half of ν_t comes from $K<p<1.1K$, whereas at $k=0.1K$ the same band contributes only 0.095 (Fig. 6). In these triads the third leg satisfies $|q|=p-k<0.11K$: energy is exchanged between two modes near the cutoff through a large-scale mode. The transfer is local in wavenumber but the interaction is nonlocal [47]. The logarithm arises because the large-scale leg has the coherent amplitude $|\hat{u}(q)|\propto1/|q|$ of a jump, so the triad weight $\propto1/[p(p-k)]$ diverges as $p\to k$. A coherent spectrum $|\hat{u}(q)|\propto|q|^{-m}$ gives a power-law divergence for $m>1$

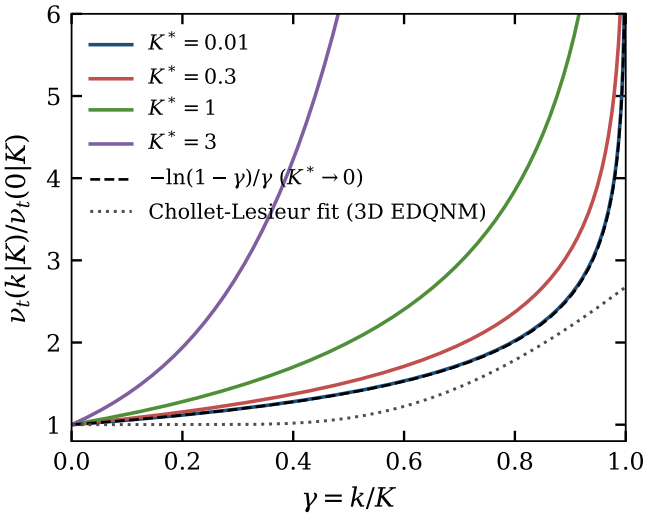


Figure 5. Exact sharp-cutoff spectral eddy viscosity of the Burgers shock, Eq. (35), normalized by its value at $k \to 0$, for $K^* = \pi\nu K/u_0 = 0.01$ to 3. Dashed: the inertial limit $\Lambda = -\ln(1-\zeta)/\zeta$, Eq. (36). Dotted: the normalized three-dimensional EDQNM fit of Chollet and Lesieur, for shape comparison only.

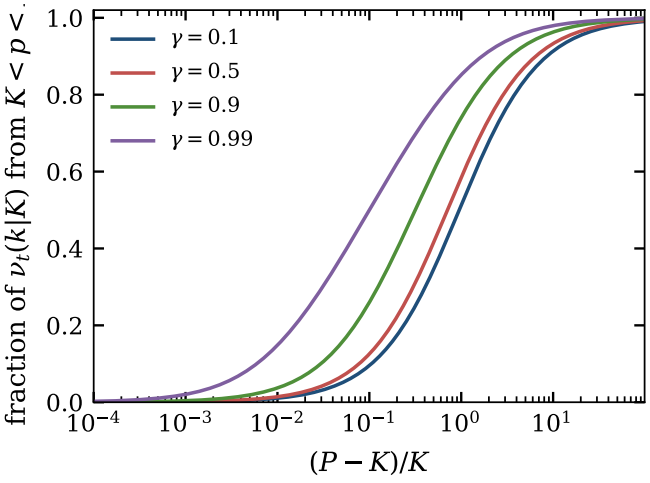


Figure 6. Cumulative fraction $F(P)$, Eq. (38), of the inertial sharp-cutoff eddy viscosity contributed by subgrid modes $K < p < P$, for $k/K = 0.1$, 0.5, 0.9 and 0.99. Near the cutoff the drain comes from a narrow band just above K, coupled to k through a large-scale mode $q = k - p$.

and a finite cusp for $m < 1$; in a random-phase closure the large-scale leg enters through $E(q)$, and the cusp is finite, consistent with the dependence of the three-dimensional cusp on the large-scale spectrum [25, 48].

7.4 Dependence on the filter, and the link to real space

For a general even filter $\hat{G}(k)$ of width Δ, the subfilter stress is $\tau = \overline{u^2} - \bar{u}^2$ and its transfer to the resolved mode k is $T_{\rm sgs}(k) = 2\pi k\mathscr{A}(k)\hat{G}(k)[\hat{G}(k)I(k) - I_G(k)]$, with $I = \int \mathscr{A}(p)\mathscr{A}(k-p)\mathrm{d}p$ and $I_G = \int \hat{G}(p)\hat{G}(k-p)\mathscr{A}(p)\mathscr{A}(k-p)\,\mathrm{d}p$. Dividing by $-2k^2\hat{G}^2E$ and taking the inertial limit, in which $I \to 0$ for $k \neq 0$,

$$\nu_t(k) = \frac{\pi}{2u_0\hat{G}(k)}\,\mathrm{P}\!\int_{-\infty}^{\infty} \hat{G}(p)\hat{G}(k-p)\left(\frac{u_0}{\pi}\right)^2 \frac{\mathrm{d}p}{p(k-p)}. \tag{39}$$

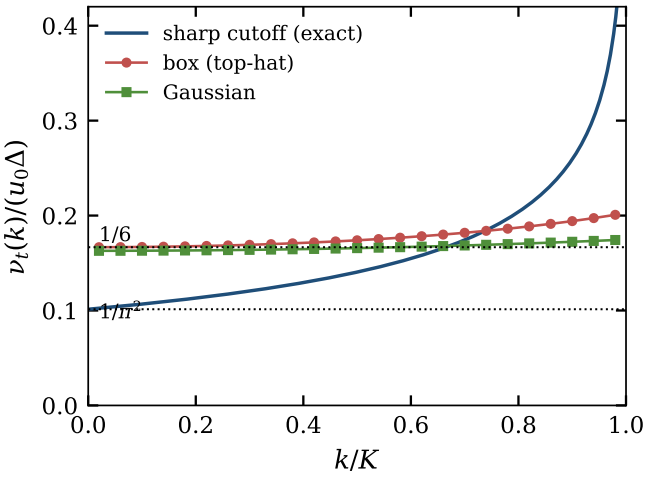


Figure 7. Inertial spectral eddy viscosity of the shock, in units of $u_0\Delta$, for a sharp cutoff (Eq. 36; $\Delta = \pi/K$), a top-hat filter and a Gaussian filter (Eq. 39, evaluated numerically). Dotted: long-wave limits $1/\pi^2$ and $1/6$, Eq. (41). Smooth filters remove the cusp.

For the sharp cutoff this is Eq. (36). Expanding $\hat{G}(k-p)$ to first order in k gives the long-wave limit for any filter (Appendix B),

$$\nu_t(k \to 0) = -\frac{u_0}{\pi}\int_0^\infty \frac{1}{p}\frac{\mathrm{d}\hat{G}^2}{\mathrm{d}p}\,\mathrm{d}p, \tag{40}$$

so that

$$\nu_t(0) = \begin{cases} u_0\Delta/\pi^2, & \text{sharp cutoff, } \Delta = \pi/K, \\ u_0\Delta/6, & \text{top hat, } \hat{G} = \mathrm{sinc}(k\Delta/2), \\ (3/\pi)^{1/2}\,u_0\Delta/6, & \text{Gaussian, } \hat{G} = e^{-k^2\Delta^2/24}. \end{cases} \tag{41}$$

The top-hat value uses $\int_0^\infty s^{-1}\mathrm{d}(\mathrm{sinc}^2 s) = -\pi/3$; the Gaussian has the same second moment as the top hat. Numerical evaluation of Eq. (39) gives $0.5236\,u_0$ and $0.5117\,u_0$ at $k\Delta = 0.06$ for $\Delta = \pi$, against $\pi/6 = 0.5236$ and 0.5117. The sharp-cutoff eddy viscosity, despite its cusp, has the same band average as the top hat: since $\int_0^1 \Lambda\,\mathrm{d}\zeta = \pi^2/6$,

$$\frac{1}{K}\int_0^K \nu_t(k|K)\,\mathrm{d}k = \frac{u_0}{\pi K}\frac{\pi^2}{6} = \frac{u_0\Delta}{6}, \qquad \Delta = \frac{\pi}{K}. \tag{42}$$

Smooth filters remove the cusp (Fig. 7): for the top hat $\nu_t(k)$ rises from $u_0\Delta/6$ by 21% at $k = 0.98K$, for the Gaussian by 7%. A smooth filter does not cut the triad integral at a sharp boundary, so the lower limit never approaches the singularity at $p = k$. Oberai *et al.* [36] found, for exponential filters normalized differently, eddy viscosities much larger than for the sharp cutoff; with widths matched to the same second moment, the increase is a factor $(\pi^2/6)(3/\pi)^{1/2} = 1.6$ for the Gaussian.

The top-hat value closes the loop with Section 6. With $\Delta = R$, the long-wave spectral eddy viscosity $u_0R/6$ equals the inertial real-space eddy viscosity of the shock obtained independently from the exact budget, Eq. (28), and from the homogenized cell, Eq. (29). Three routes (exact budget, cell problem, triad integral) thus give the same eddy viscosity for the same filter. For a top-hat filter a single wavenumber-independent eddy viscosity describes the drain of a shock to within 21% at all resolved wavenumbers, which supports Smagorinsky-type models for top-hat-filtered Burgers simulations [49, 50].

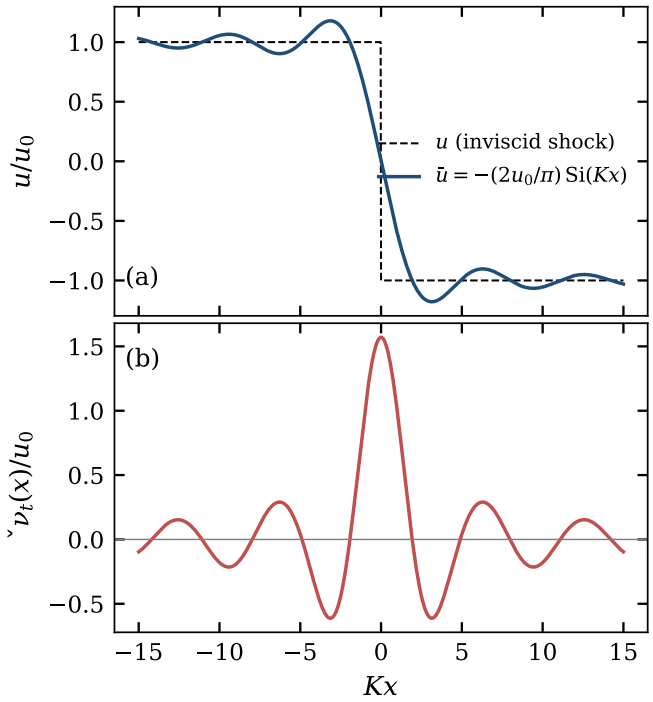


Figure 8. (a) Inviscid shock and its sharp-filtered profile $\bar{u} = -(2u_0/\pi)\mathrm{Si}(Kx)$. (b) Real-space kernel of the sharp-cutoff spectral eddy viscosity, Eq. (43): proportional to the subgrid kinetic-energy density, with a central peak at the shock and negative lobes where $|\bar{u}| > u_0$.

7.5 Real-space kernel of the sharp-cutoff eddy viscosity

The sharp-filtered inviscid shock is $\bar{u} = -(2u_0/\pi)\,\mathrm{Si}(Kx)$. Its square has the transform $\mathscr{F}(\bar{u}^2) = u_0^2\delta(k) + \frac{2u_0^2}{\pi^2|k|}\ln|1-|k|/K|$ for $0 < |k| < 2K$, while $\mathscr{F}(u^2) = u_0^2\delta(k)$. Comparing with Eq. (36), $\nu_t(k|K) = \frac{\pi}{2u_0}\mathscr{F}(u^2-\bar{u}^2)(k)$ for $0 < |k| < 2K$, so the kernel $\check{\nu}_t(x) = \int \nu_t(k)e^{-ikx}\mathrm{d}k$ is

$$\check{\nu}_t(x) = \frac{\pi}{2u_0}\left[u^2-\bar{u}^2\right] = \frac{\pi}{u_0}\,e_{\mathrm{sgs}}(x), \qquad e_{\mathrm{sgs}} = \tfrac{1}{2}\left(u^2-\bar{u}^2\right): \quad (43)$$

up to the factor π/u_0, the subgrid kinetic-energy density. The subgrid force is $\frac{1}{2\pi}\int \check{\nu}_t(x-x')\,\partial_{x'}^2\bar{u}\,\mathrm{d}x'$. The kernel (Fig. 8) has a central peak $\pi u_0/2$ at the shock, negative side lobes where $|\bar{u}|$ overshoots u_0 (Gibbs oscillations), and integral $2\pi\nu_t(0|K) = 2u_0/K$. The cosine transform of Eq. (36) and Eq. (43) agree to six digits. A sharp cutoff thus implies a nonlocal, sign-changing eddy viscosity in real space although $\nu_t(k|K) > 0$ at every wavenumber.

7.6 Ensembles of shocks and decay

Burgers turbulence at high Reynolds number is a gas of shocks separated by ramps [39, 51]. For shocks with half-jumps u_{0j} at independent, uniformly distributed positions x_j, the phase factor $e^{-ikx_j+ipx_l+i(k-p)x_m}$ of a triple product averages to zero unless $j = l = m$, so transfer and spectrum are sums over shocks. With each $T_j \propto u_{0j}^3$ and $E_j \propto u_{0j}^2$, the inertial ensemble eddy viscosity is

$$\nu_t(k|K) = \frac{\langle u_0^3\rangle}{\langle u_0^2\rangle}\frac{1}{\pi K}\Lambda(\zeta), \quad (44)$$

and at finite viscosity each shock contributes Eq. (35) with weight $E_j(k) \propto \sinh^{-2}(\pi\nu k/u_{0j})$. The shape, the cusp and the filter dependence are unchanged; the plateau is weighted toward the strongest shocks. The ramps carry the large-scale velocity and provide the large-scale legs of Section 7.3. When the large-scale spectrum $E \propto k^n$ is preserved during decay, $u_0 \propto t^{-(n+1)/(n+3)}$ and $\ell \propto t^{2/(n+3)}$ [39, 51]; at fixed cutoff the plateau decays as u_0, and for a single sawtooth of period ℓ, $u_0 = \ell/(2t)$ and $\nu_t^+ \propto t^{-1}$. Section 11.2 tests Eq. (35) with shock-weighting against DNS.

8 THE INERTIAL RANGE: A DERIVATION OF THE TWO-THIRDS LAW

8.1 Scale invariance makes the eddy viscosity linear

In the inertial range the subcell velocity of a cell of size R contains the eddies of size R and smaller. Suppose its statistics are self-similar, with $\langle[v(y)-v(0)]^2\rangle \propto y^{2H}$ inside the cell. The only velocity scale is $S_2(R)^{1/2}$ and the only dimensionless parameter is the cell Reynolds number $Re_R = RS_2(R)^{1/2}/\nu$. The displacement is $\xi(y) = Re_R\,\Phi(y)$, where $\Phi = \int_0^y (B_H - \bar{B}_H)\mathrm{d}y'$ is a universal random profile built from a unit-amplitude process B_H with exponent H. For $Re_R \gg 1$, Eq. (18) gives

$$\langle\nu_e\rangle = c\,Re_R - \frac{1}{1+H} + o(1), \qquad c = \left\langle \Phi_{\max} - \textstyle\int_0^1 \Phi\,\mathrm{d}y\right\rangle, \quad (45)$$

where the offset uses $\alpha = 1+H$ (Φ is one degree smoother than B_H). In dimensional form

$$\nu_t(R) = c\,RS_2(R)^{1/2}. \quad (46)$$

This is the mixing-length form of the eddy viscosity [52, 53]; here it is the advective limit of the cell solution under A1, valid for every H because the cell problem is linear in ξ at large Re_R, and it comes with its coefficient and with its breakdown at small Re_R, where $\langle\nu_e\rangle = aRe_R^2$ with $a = \langle\int\Phi^2 - (\int\Phi)^2\rangle$.

Monte Carlo sampling of Gaussian random-phase cells with stationary increments (Cholesky factorization of the increment covariance on 512–4096 points; 84 000 samples for $H = 1/3$) gives $c = 0.0817 \pm 0.0004$ for $H = 1/3$ (no trend with the grid size), $c = 0.0738$, 0.0627 and 0.0502 for $H = 1/2$, $3/4$ and 1, and $a = 0.00315$ for $H = 1/3$. Figure 9(a) shows the curves $\langle\nu_e\rangle(Re_R)$ with both asymptotes; at $Re_R = 3000$ the computed mean and Eq. (45) agree to four digits, including the offset.

8.2 The exact budget selects $H = 1/3$

Insert Eq. (46) into Eq. (6) with $\nu_e \gg 1$:

$$c\,RS_2^{1/2}\frac{S_2}{R^2} = \varepsilon \quad \Rightarrow \quad S_2(R) = \left(\frac{\varepsilon R}{c}\right)^{2/3}. \quad (47)$$

This is the two-thirds law. The exponent comes from the exact identity $\Omega_R = S_2/R^2$ and from the linearity of the advective cell, and the constant is $C_2 = c^{-2/3} = 5.31$ for one-dimensional random-phase cells. The argument is self-consistent: the cell statistics were assumed self-similar with unknown H, and the budget holds only if $2H = 2/3$; any other H would give a flux $\nu_t\Omega_R \propto R^{3H-1}$ that changes with scale. Equivalently, with $\nu_t \propto R^{4/3}$ the homogenized equation $\partial_t\bar{u} + \bar{u}\cdot\nabla\bar{u} = -\nabla\bar{p} + \nabla\cdot(\nu_t\nabla\bar{u})$ is invariant under $x \to \lambda x$, $\bar{u} \to \lambda^{1/3}\bar{u}$, $t \to \lambda^{2/3}t$. The exponent does not use Gaussianity, the value of c or the separable reduction of Section 9; it does use A1 through the linear form Eq. (46). The cell eddy viscosity equals the molecular viscosity at $Re_R \approx 21$ (Fig. 9a).

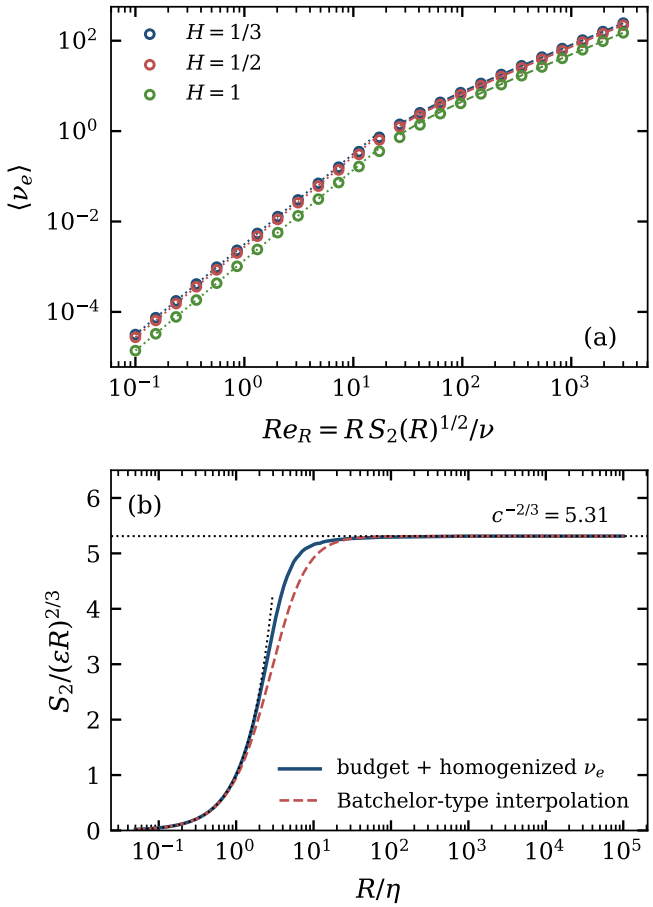


Figure 9. (a) Mean cell eddy viscosity of self-similar Gaussian cells against $Re_R = RS_2^{1/2}/\nu$ for $H = 1/3$, $1/2$ and 1 (symbols, Monte Carlo); dashed lines $cRe_R - 1/(1+H)$, dotted lines aRe_R^2. (b) Structure function from the exact one-dimensional budget, Eq. (6), closed with the $H = 1/3$ cell eddy viscosity, compensated by $(\varepsilon R)^{2/3}$; dotted: the dissipative law $(R/\eta)^{4/3}$ and the plateau $C_2 = c^{-2/3}$; dashed: a Batchelor-type interpolation with the same limits.

8.3 The full structure function

Equation (6) with the computed $\langle \nu_e \rangle (Re_R)$ holds at every scale. In Kolmogorov units, $x = R/\eta$ and $s_2 = S_2/u_\eta^2$, it reads $[1 + \langle \nu_e \rangle (x\sqrt{s_2})]\, s_2 = x^2$. Its solution (Fig. 9b) goes from the dissipative law $s_2 = x^2$ to $s_2 = C_2 x^{2/3}$, with the crossover at $R_c = c^{-1/2}\eta = 3.50\,\eta$. The shape is close to the interpolation $s_2 = x^2/[1 + (x/R_c)^2]^{2/3}$ [54], with a sharper transition because the cell eddy viscosity switches from quadratic to linear growth within a decade of Re_R. In the dissipative range $\nu_e \ll 1$ and the result is insensitive to the closure there, including the omission of gradient transfer.

9 THREE DIMENSIONS: AN ESTIMATE OF THE KOLMOGOROV CONSTANT

The budget Eq. (4) with Eq. (7) holds in three dimensions; the cell problem Eq. (12) does not have a closed-form solution there. We use a separable approximation (assumption A3): in each Cartesian direction the cell is a one-dimensional cell advected by the longitudinal velocity component; transverse advection and pressure inside the cell are neglected. By isotropy,

$$\nu_t(R) = cRS_{2L}(R)^{1/2}, \tag{48}$$

with S_{2L} the longitudinal structure function. For $E \propto k^{-p}$, $S_{2L} \propto R^{p-1}$ and $\Omega_R \propto R^{p-3}$, so the flux $\nu_t \Omega_R \propto R^{(3p-5)/2}$ is scale-independent only for $p = 5/3$. With $E = C_K \varepsilon^{2/3} k^{-5/3}$, $S_{2L} = C_{2L}(\varepsilon R)^{2/3}$ with $C_{2L}/C_K = \frac{27}{55}\Gamma(\frac{1}{3}) = 1.3151$ [4], and the budget $\nu_t \Omega_R = \varepsilon$ gives

$$C_K = \left[2c\,(C_{2L}/C_K)^{1/2} I\right]^{-2/3}, \tag{49}$$

where I is the filter integral of Eq. (7). With $c = 0.0817$ this yields $C_K = 1.36$ for the cube cell, the natural geometry of a separable top-hat filter, and 1.10 for a ball of diameter R; the longitudinal constants are $C_{2L} = 1.79$ and 1.44. Measured values are $C_K \approx 1.5$–1.6 and $C_{2L} \approx 2.0$ [5, 6, 7]. The cube estimate is 10–15% low; the Lagrangian renormalized approximation, 1.72, is about 10% high [18]. The spread between cube and ball measures the geometric ambiguity in identifying the cell with the filter.

Writing $\nu_t = C_S^2 R^2 \Omega_R^{1/2}$ [20] and $\Omega_R = \kappa S_{2L}/R^2$ with $\kappa = 2I/(C_{2L}/C_K)$ ($\kappa = 5.13$ for the cube), Eq. (48) gives $C_S = (c/\kappa^{1/2})^{1/2} = 0.190$ (cube) or 0.175 (ball). Lilly's estimate is $C_S \approx 0.17$, and values of 0.1–0.2 are used in practice [4, 21].

Three approximations limit Eq. (49): (i) Gaussian random-phase cells, whereas the velocity increments of turbulence are negatively skewed; (ii) the separable reduction A3; (iii) the identification of the cell with the filter volume, exact in one dimension but approximate in three. Section 11.3 measures each of them: (i) changes c by a few per cent, while (ii) together with A1 reduces the transfer by 10–35%. The exponent does not depend on any of them.

9.1 Spectral form and the two-dimensional enstrophy range

With a sharp spectral cutoff the stationary budget reads $2\nu[1 + \hat{\nu}(K)] \int_0^K Ek^2 \mathrm{d}k = \varepsilon$, so $\hat{\nu}(K) = \int_K^\infty Ek^2 \mathrm{d}k / \int_0^K Ek^2 \mathrm{d}k$, the definition used by Kraichnan [22]. Closing it with $\nu_t \propto K^{-1}[KE(K)]^{1/2}$, a spectrum $E \propto K^{-p}$ gives an energy flux $\propto K^{(5-3p)/2}$, constant only for $p = 5/3$. In the direct cascade of two-dimensional turbulence the budget is weighted by k^4: $2\nu[1 + \hat{\nu}_2(K)] \int_0^K Ek^4 \mathrm{d}k = \eta_Z$, the enstrophy dissipation rate; the same closure gives an enstrophy flux $\propto K^{(9-3p)/2}$, constant for $p = 3$ [55]. The logarithmic correction of that range is not captured by the local closure.

10 PHASE COHERENCE AND INTERMITTENCY

Equation (46) holds when the cell statistics are random-phase and self-similar, so that $\xi_{\max}$ scales with $RS_2^{1/2}$. If a cell contains a coherent structure whose amplitude is not set by the rms increment, the cell eddy viscosity depends on the outer scale ℓ,

$$\nu_t = cRS_2(R)^{1/2} \left(\frac{\ell}{R}\right)^p, \tag{50}$$

and the exact budget gives

$$S_2(R) = \left(\frac{\varepsilon R}{c}\right)^{2/3} \left(\frac{R}{\ell}\right)^{2p/3}, \tag{51}$$

the β-model correction $(R/\ell)^{(d-D)/3}$ of Frisch *et al.* [37] with fractal dimension $D = d - 2p$ of the active region. The Burgers equation is the extreme case. For a dilute train of shocks of half-jump u_0 and spacing ℓ, a fraction R/ℓ of the cells contains a shock, so

$S_2 \simeq 4u_0^2 R/\ell$, while Section 6 gave $\nu_t = u_0 R/6$ for those cells. Hence $\nu_t = \frac{1}{12} R S_2^{1/2} (\ell/R)^{1/2}$: $p = 1/2$, $D = 0$, $S_2 \propto R$ and $E \propto k^{-2}$, the known Burgers results [39, 51]. The budget is satisfied exactly: $\nu_t S_2/R^2 = \frac{2}{3} u_0^3/\ell = \varepsilon$. In spectral form $E(k) \simeq 2u_0^2/(\pi \ell k^2) = C_B \varepsilon^{2/3} \ell^{-1/3} k^{-2}$ with $C_B = (2/\pi)(3/2)^{2/3} = 0.834$, for a steady and for a decaying train of shocks at any instant. The same ℓ-dependence appears in the spectral plateau Eq. (37). In this picture intermittency is a statement about phase coherence inside the cell: random phases give $p = 0$ and K41, a single coherent structure per active cell gives $p = (d - D)/2$. For three-dimensional turbulence the exponent p is an input, not a result.

11 TESTS AGAINST DIRECT NUMERICAL SIMULATION

11.1 Data and procedure

Burgers turbulence. We use decaying solutions of the Burgers equation on a 2π-periodic domain with $N = 32\,768$ grid points ($\Delta x = 1.92 \times 10^{-4}$), $\nu = 5 \times 10^{-4}$ and time step 10^{-5}, ten independent realizations and twenty stored snapshots. We analyse snapshots 5, 10 and 19, which contain on average 7.8, 8.7 and 6.5 shocks per period (mean spacing $\ell = 0.81$, 0.72 and 0.97) and have $u_{\rm rms} = 0.40$, 0.30 and 0.23. The shocks are resolved: their thickness ν/u_0 is about six grid spacings. A shock is a local minimum of u_x below $-0.2 \max|u_x|$; its half-jump u_0 is half the difference between the maximum of u in the 0.01 before it and the minimum in the 0.01 after it.

Isotropic turbulence. We use the forced isotropic turbulence of the JHTDB [34, 35]: a pseudo-spectral DNS on a 1024^3 periodic grid of side 2π with $\nu = 1.85 \times 10^{-4}$, time-averaged $\varepsilon = 0.103$, $\eta = 0.0028$ and $R_\lambda \approx 420$, forced at wavenumbers $|\mathbf{k}| \leqslant 2$. Two velocity cubes of 256^3 points at the last stored time were used: (C1) a contiguous subcube at the grid spacing $\Delta x = 2\pi/1024 = 2.19\eta$, of side $\pi/2$; and (C4) the whole periodic domain sampled at every fourth grid point (spacing 8.8η). As a check we also used a contiguous 256^3 subcube of the JHTDB `isotropic4096` data set at its grid spacing.

Procedure. For a cube (top-hat) filter of m grid points, $R = mh$ with h the spacing, we compute $\bar{u}_i$ and $\overline{u_i u_j}$ by running sums, the resolved gradients by central differences, Ω_R, $\Pi_R = -\langle \tau_{ij} \bar{S}_{ij} \rangle$, the measured eddy viscosity $\nu_t = \Pi_R/\Omega_R$ (the definition Eq. 3), and $S_{2L}(R)$ averaged over the three axes; on C1 averages exclude points whose filter or difference stencil leaves the cube, on C4 the data are periodic. For Burgers the same quantities follow from Eqs. (3)–(5) with $\Pi_R = -\frac{1}{2} \langle \tau \partial_x \bar{u} \rangle$, the form for the Burgers nonlinearity $\partial_x(u^2/2)$. The predictions use no fitted parameter: the cell eddy viscosity Eq. (16) is evaluated either from the Gaussian cells of Section 8 or directly from DNS velocity segments, and the spectral prediction from the shocks detected in each snapshot.

11.2 Burgers turbulence

Real space. For every cell $[x_0, x_0 + R]$ we evaluate Eq. (16) with $v = (R/\nu)(u - \bar{u}_{\rm cell})$ and average with the weight $[u(x_0 + R) - u(x_0)]^2$, as in Section 6. Figure 10(a) compares the result with the measured ν_t/ν. For $R = 128$–$512\Delta x$ ($R/\ell \approx 0.03$–0.12), where ν_t/ν grows from about 2 to 14, measured/predicted is 0.95–1.29 for the three snapshots: the closure, built on a single steady shock, describes the drain of a decaying shock gas. At small R the measured $\nu_t \propto R^2$ exceeds the prediction $\propto R^4$ by orders of magnitude: this is the gradient (Leonard) transfer of Section 6, which the closure omits. At $R/\ell \approx 0.2$–0.5 the ratio falls to 0.39–0.94: cells comparable with the shock spacing contain ramps and neighbouring shocks, and treating each cell independently overestimates the drain.

Spectral space. We measure $\nu_t(k|K)/\nu = -\sum \mathrm{Re}[\hat{u}^*(\hat{N}(u) - \hat{N}(u^<))]/\sum \nu k^2 |\hat{u}|^2$ for a sharp cutoff, with $N(u) = -\partial_x(u^2/2)$ and $u^<$ the field truncated at K, summing over realizations and over 25 bins in k. The prediction sums Eq. (35) over the detected shocks with weights $\sinh^{-2}(\pi \nu k/u_{0j})$ (Section 7.6); the intensity-weighted half-jumps are $\langle u_0^3 \rangle / \langle u_0^2 \rangle = 0.63$, 0.44 and 0.35. Figure 10(b) shows snapshot 10. For $K = 128$ and 512, which put the cutoff at $K^* \approx 0.3$–2.3, the median measured/predicted ratio over $0.1 < k/K < 0.95$ is 0.82–0.91 for the three snapshots, and the rise toward the cutoff is reproduced. At $K = 32$ the ratio scatters more because few modes fall in each bin and the cutoff wavelength spans a large fraction of the shock spacing, so triads coupling different shocks no longer average out. In the last bin before K the measured cusp is lower than predicted, as expected when the logarithmic singularity is averaged over a finite bin and the large-scale legs are ramps rather than isolated jumps. The prediction, from the measured shock amplitudes alone, is systematically 9–22% high; the shape is right.

11.3 Isotropic turbulence

The exact budget. At $R = \Delta x$ on C1, $\nu\Omega_R = 0.093$ (central differences on a 2.2η grid underestimate the gradient variance), and for $R \leqslant 16\Delta x$ (35η) the sum $\nu\Omega_R + \Pi_R$ lies between 0.093 and 0.113, close to the time-averaged $\varepsilon = 0.103$. At larger R the subcube, of the size of the integral scale, gives up to 0.132, a sampling fluctuation of a single region. On the full domain C4 the sum is 0.085–0.091 for $R = 35$–280η, a nearly constant flux slightly below ε because the subsampled field misses part of the gradient variance. Within these uncertainties, about 10%, Eq. (4) holds at all scales, as it must.

Magnitude of the eddy viscosity. Figure 11(a) shows the measured ν_t/ν against $Re_R = R S_{2L}^{1/2}/\nu$ with the Gaussian-cell curve of Section 8 and the one-dimensional cell eddy viscosity evaluated on 30 000 segments of the longitudinal DNS velocity (refined eightfold, sixteenfold for C4, by linear interpolation). Above $R \approx 18\eta$ the prediction follows the data over two decades of ν_t/ν. At $R = 18\,\eta$ ($Re_R = 57$) the measured value is 2.66 and the DNS-cell value 2.62. In the inertial range, $R = 35$–140η, measured/DNS-cell is 0.74–0.90: the separable closure overestimates the eddy viscosity by 10–35%. At $R = 280\eta$, about 0.6 of the integral scale, the ratio is 0.67. Below 10η the measured eddy viscosity is larger than the cell value (0.20 against 0.03 at $4.4\,\eta$, 0.81 against 0.39 at $8.8\,\eta$): gradient transfer again.

The coefficient c. Figure 11(b) shows $\nu_t/(R S_{2L}^{1/2})$. On the full domain it is flat, $c = 0.060$, at $R = 70$–140η; on the subcube it rises from 0.047 to 0.067 between 18η and 140η. The one-dimensional cells built from DNS segments give $c = 0.074$–0.083 at $R \geqslant 70\eta$, within 0.93–1.03 of the Gaussian-cell curve at the same Re_R. The `isotropic4096` subcube gives $c = 0.061$ at $R = 32$ grid spacings, where its flux has not reached a plateau.

Gaussianity and skewness. The DNS segments have increment skewness -0.41 to -0.23. Reflecting every segment, $v \to -v$, which exchanges converging and diverging zones and reverses the skewness, lowers the cell eddy viscosity by 1–4%. Negative skewness therefore raises the cell eddy viscosity, as anticipated in Section 9, but only by a few per cent, and the near-agreement of DNS cells with Gaussian cells shows that random-phase Gaussian statistics are not the source of the discrepancy. The 10–35% overestimate comes from the dynamics omitted by A1 and A3 (pressure, transverse advection and back-reaction of the resolved field inside the cell), whose net ef-

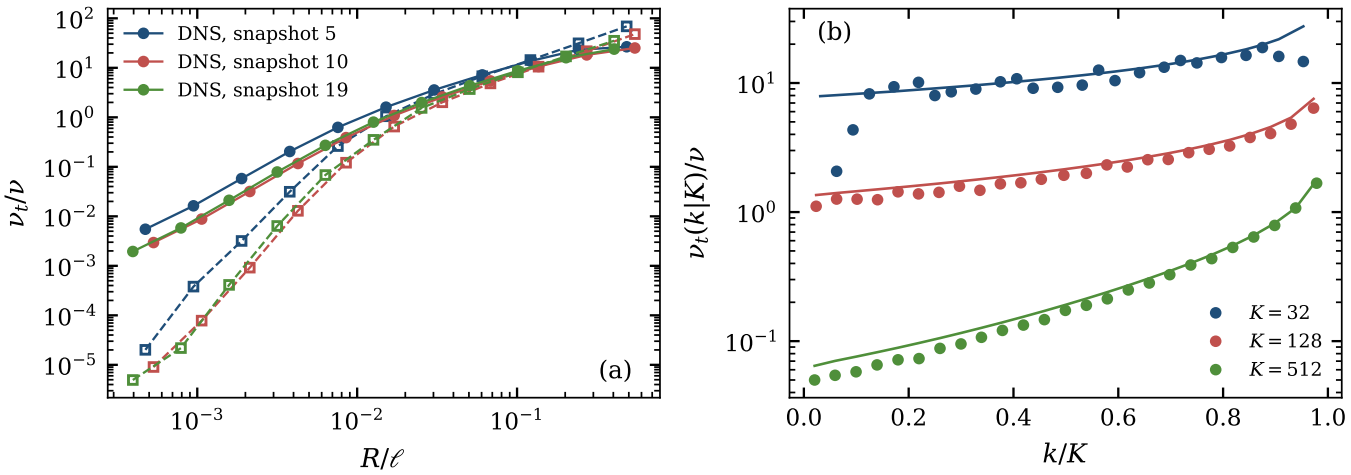


Figure 10. A priori tests in decaying Burgers turbulence. (a) Measured eddy viscosity $\nu_t = \Pi_R/\Omega_R$ (filled symbols) and homogenized cell prediction (open symbols, Eq. 16 on DNS cells, $(\Delta u)^2$-weighted) against R/ℓ for three snapshots, averaged over ten realizations. (b) Sharp-cutoff spectral eddy viscosity, snapshot 10: DNS (symbols) and the exact single-shock result Eq. (35) summed over the shocks detected in the DNS (lines), for $K = 32$, 128 and 512. No parameter is fitted.

fect is to reduce the transfer. The present test does not separate these effects.

Constants. The measured Smagorinsky coefficient on C4 is $C_S = 0.159$ at $R = 70$–140η (against 0.190 predicted; 0.152 on the `isotropic4096` subcube), and $\kappa = \Omega_R R^2/S_{2L} = 5.7$ (against 5.13 for a pure $k^{-5/3}$ spectrum). Inserting the measured $c = 0.060$ into Eq. (49) gives $C_K = 1.67$ and $C_{2L} = 2.19$, within about 10% of the measured values. The structure of the argument, the exact budget with $\nu_t = cRS_{2L}^{1/2}$, is therefore consistent with the DNS; the error of the parameter-free constant $C_K = 1.36$ is the error of the separable coefficient c.

These are a priori tests on single snapshots of one simulation at $R_\lambda \approx 420$; the inertial range spans less than a decade, and the subcube statistics fluctuate. The conclusions are robust to these limitations in the following sense: the two independent data sets (contiguous subcube and subsampled full domain) agree where they overlap ($R = 35$–140η), and the ratio of measured to cell eddy viscosity has the same sign and similar size in both.

12 DISCUSSION

12.1 The phase diagram of the cell eddy viscosity

For a subcell velocity with rms value Σ and correlation length λ the cell eddy viscosity depends on $\sigma = \Sigma R/\nu$ and $\mu = \lambda/R$ (Fig. 12). Four regimes appear: (I) advective cells with many independent eddies, $\langle \nu_e \rangle \simeq \sigma\sqrt{\pi\mu}/2$ (Eq. 24 with $s^2 = 2\sigma^2\mu$); (II) diffusive cells with many eddies, $\sigma^2\mu/6$; (III) diffusive cells smaller than an eddy, $\sigma^2/(180\mu)$; (IV) advective cells smaller than an eddy, $\simeq 0.0738\,\sigma(2/\mu)^{1/2}$, the $H = 1/2$ result of Section 8.1 for the rough exponential correlation. All four asymptotes agree with the computed map. The boundaries are $\langle \nu_e \rangle \approx 1$, $\sigma\mu^{1/2} \approx 1$ and the echo maximum $\mu \approx 0.18$. Inertial-range cells, $\mu = O(1)$ and $\sigma \gg 1$, sit in the advective half of the diagram, where the eddy viscosity is linear in σ; this is what produces the two-thirds law.

12.2 What is derived and what is assumed

Exact: the filtered energy budget Eq. (4) with the flux definition of the eddy viscosity; the one-dimensional identity $\Omega_R = S_2/R^2$; the homogenization expansion Eqs. (9)–(13); the solution Eq. (16) and its consequences; the shock results of Sections 6 and 7. Assumed: (A1) inside a cell, resolved momentum is advected by a prescribed subcell velocity, without pressure or back-reaction; (A2) a zero boundary condition on the cell faces; (A3) in three dimensions, independent longitudinal one-dimensional cells; the constants also use Gaussian random-phase cells and the identification of cell and filter volume. The exponent 2/3 needs A1, A2, self-similar cell statistics and the large-Re_R limit. A1 is a closure hypothesis; it replaces the mixing-length hypothesis by a solvable cell problem, but it is not a consequence of the Navier–Stokes equations, and the result is a derivation of K41 *within* this closure.

12.3 Strengths and weaknesses

Strengths.

(i) Every assumption is explicit, and the exact parts are separated from the modelled ones. The budget that converts an eddy viscosity into $S_2(R)$ is exact, and the DNS confirm it in three dimensions to about 10%.

(ii) The one-dimensional cell problem is solved exactly, with positivity, the offset rule, a closed form for Brownian cells and the stochastic echo; these results are independent of turbulence and apply to any one-dimensional homogenization problem of this type.

(iii) The mixing-length form $\nu_t = cRS_2^{1/2}$ is obtained as the advective limit of a solved problem, together with its coefficient and its diffusive limit. The exponent 2/3 and the selection of $H = 1/3$ follow without dimensional analysis and without Gaussianity.

(iv) The closure has no adjustable parameter. It gives $C_K = 1.36$, $C_S = 0.190$ and the shape of S_2 across the dissipative and inertial ranges.

(v) For the Burgers shock three independent routes (exact budget, cell problem, triad integral with a top-hat filter) give the same eddy viscosity $u_0R/6$, and the exact spectral results (finite-viscosity form, triad origin of the cusp, general-filter formula, band average, real-

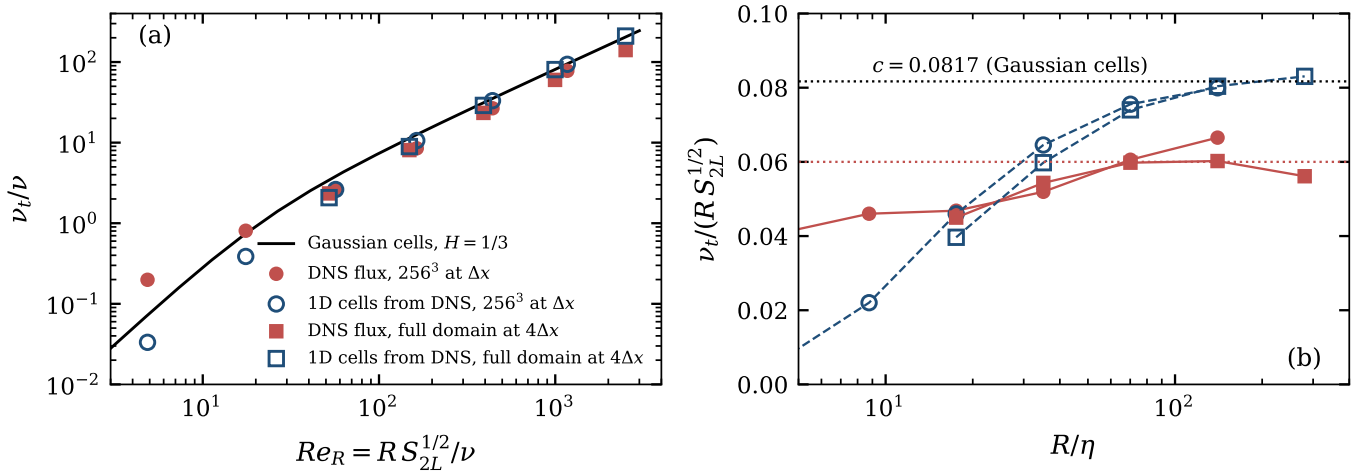


Figure 11. A priori test in forced isotropic turbulence (JHTDB, $R_\lambda \approx 420$). (a) Eddy viscosity against Re_R: measured flux-based $\nu_t = \Pi_R/\Omega_R$ (filled), one-dimensional cell eddy viscosity evaluated on longitudinal DNS velocity segments (open), and the Gaussian $H = 1/3$ cell curve of Section 8 (line). Circles: contiguous 256^3 subcube at the grid spacing; squares: whole domain sampled at every fourth point. (b) Coefficient $\nu_t/(RS_{2L}^{1/2})$ against R/η: DNS flux (filled, solid) and DNS cells (open, dashed); dotted lines mark the Gaussian-cell value $c = 0.0817$ and the measured plateau 0.060.

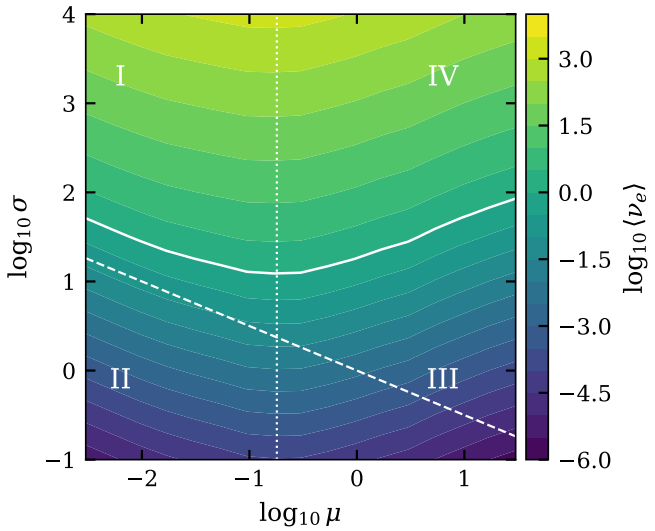


Figure 12. Mean cell eddy viscosity (colour, $\log_{10}$) for pinned cells with Ornstein–Uhlenbeck subcell velocity against $\mu = \lambda/R$ and $\sigma = \Sigma R/\nu$. White solid line: $\langle\nu_e\rangle = 1$; dashed: $\sigma\mu^{1/2} = 1$; dotted: echo maximum μ^*. Roman numerals mark the four regimes of Section 12.1.

space kernel, ensembles) form a benchmark for spectral closures and subgrid models.

(vi) The predictions survive a priori DNS tests without fitted parameters: Burgers eddy viscosities to within 30% in real space and 9–22% in spectral space; in isotropic turbulence, one-dimensional cells built from DNS velocities reproduce the Gaussian coefficient within 7% at $R \geqslant 70\eta$, and the predicted eddy viscosity is within 10–35% of the measured one for R above about 18η.

Weaknesses.

(i) A1 is a closure. The DNS show that A1 and A3 together overestimate the three-dimensional eddy viscosity by 10–35% in the inertial range ($c = 0.0817$ against 0.060); this is why the parameter-free $C_K = 1.36$ is 10–15% low.

(ii) The cell eddy viscosity is never negative, so the closure has no backscatter. Multiscale expansions that keep pressure and back-reaction can give negative eddy viscosities [32, 33], and local backscatter is well documented in turbulence [56].

(iii) The closure omits gradient (Leonard) transfer. Its cell eddy viscosity grows as R^4 in the dissipative range, whereas the exact shock value and both DNS grow as R^2; the spectral form of the budget then gives $E \propto k^{-7}$ far beyond the Kolmogorov wavenumber (Heisenberg's result [9]) instead of the observed exponential decay [7]. The theory is designed for the inertial range and the inertial–dissipative crossover of S_2, where ν_e is either large or irrelevant.

(iv) The identification of cell and filter volume is ambiguous in three dimensions (cube $C_K = 1.36$, ball 1.10).

(v) In three dimensions intermittency enters through the exponent p of Eq. (50), which is an input; only for Burgers is p derived.

(vi) The spectral results are exact for coherent shocks only. Their logarithmic cusp is stronger than the finite cusp of three-dimensional closures, and their inviscid sharp-cutoff limit was known [36].

(vii) The DNS tests are a priori, on single snapshots of one isotropic simulation with a short inertial range, and on one set of decaying Burgers runs; they do not test the closure inside an LES.

12.4 Relation to earlier closures and further tests

Equation (46) has the form of Prandtl's mixing length and of the eddy viscosities of Heisenberg, Obukhov and Kovasznay [9, 10, 52]. Here it is the advective limit of a solved cell problem with a computed coefficient and a computed breakdown at small cell Reynolds number. Compared with Lagrangian closures, which obtain $C_K \approx 1.72$ [18], the present closure is cruder in its dynamics but simpler: it reduces the cascade to a one-dimensional exactly solvable problem plus an exact budget, and the DNS locate its error in a single coefficient. In the random-walk picture of the cascade [16], the waiting time of energy at scale R is R^2/ν_t; Eq. (46) identifies it with the eddy turnover time $R/S_2^{1/2}$ divided by c.

The a priori procedure of Section 11 follows the standard practice for subgrid models [21, 46, 56, 57]. Three further tests follow from the theory: the cell eddy viscosity conditioned on the position of the maximum of the integrated longitudinal velocity should follow Eq. (18); a three-dimensional cell problem with DNS subcell

velocities, with and without pressure, would measure the separate contributions of transverse advection, pressure and back-reaction; and data at higher R_λ would test whether c stays at 0.060.

13 CONCLUSIONS

Two-scale stochastic homogenization of the resolved momentum, combined with the exact energy budget of the top-hat-filtered Navier–Stokes equations, gives a parameter-free eddy viscosity and, through it, the two-thirds law. In one dimension the budget is exact and the cell problem is solved exactly; its eddy viscosity is quadratic in the cell Reynolds number when diffusion dominates and linear when advection dominates, $\nu_t = cRS_2^{1/2}$ with $c = 0.0817$ for random-phase cells, and the budget then gives $S_2 \propto (\varepsilon R)^{2/3}$ and selects the Hölder exponent $1/3$. For the Burgers shock the closure, the exact budget and the exact triad integral with a top-hat filter all give $u_0 R/6$; the sharp-cutoff spectral eddy viscosity of the shock is known in closed form at finite viscosity, its cusp comes from nonlocal triads, and its long-wave value for any filter follows from one integral. A priori tests confirm the exact budget in three-dimensional DNS, confirm the shock predictions in Burgers DNS to within 30%, and show that one-dimensional cells built from DNS velocities reproduce the Gaussian coefficient. They also show the main limitation: the separable closure overestimates the three-dimensional eddy viscosity by 10–35%; with the measured coefficient $c = 0.060$ the budget gives $C_K \approx 1.67$. A three-dimensional cell problem that keeps pressure and transverse advection is the next step toward a parameter-free Kolmogorov constant.

AUTHOR CONTRIBUTIONS

Z.J. Xu performed the conception, analysis, and writing.

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DATA AVAILABILITY

The data behind every figure are archived on Zenodo, doi:10.5281/zenodo.24096071 [58]. The isotropic-turbulence data are publicly available from the Johns Hopkins Turbulence Databases. The Burgers simulation data and the Python scripts that produce every number and figure are available from the author on request.

APPENDIX A: MEAN EDDY VISCOSITY OF A FREE BROWNIAN CELL

Let $\xi = sW(y)$ with W a standard Brownian motion on $[0,1]$, $Z = \int_0^1 e^{\xi}\,\mathrm{d}y$ and $\beta^2 = s^2/2$. By Eq. (17) and $\langle \int \xi \rangle = 0$, $\langle \nu_e \rangle = s\,\mathrm{d}\langle \ln Z\rangle/\mathrm{d}s = \beta\,\mathrm{d}\langle \ln Z\rangle/\mathrm{d}\beta$.

Moments. For $y_1 > y_2 > \cdots > y_n$, $\langle \exp(s\sum_i W(y_i))\rangle = \exp[\beta^2 \sum_i (2i-1)y_i]$. Integrating over the ordered simplex, multiplying by $n!$ and using $\sinh^{2n} x = 2^{1-2n}[\sum_{i=0}^{n-1}(-1)^i \binom{2n}{i}\cosh(2(n-i)x) + \frac{1}{2}(-1)^n\binom{2n}{n}]$ and $e^{m^2\beta^2} = \frac{2}{\beta\sqrt{\pi}}\int_0^\infty e^{-x^2/\beta^2}\cosh(2mx)\mathrm{d}x$,

$$\langle Z^n \rangle = \frac{2}{\beta^{2n+1}\Gamma(n+\frac{1}{2})}\int_0^\infty e^{-x^2/\beta^2}\sinh^{2n} x\,\mathrm{d}x. \tag{A1}$$

For $n = 1$ this is $(e^{\beta^2}-1)/\beta^2$, the exact mean of Z.

Laplace transform. Summing $\phi(p) = \langle e^{-pZ}\rangle = \sum_n (-p)^n \langle Z^n\rangle/n!$ with $n!\,\Gamma(n+\frac{1}{2}) = \sqrt{\pi}\,(2n)!/4^n$,

$$\phi = \frac{2}{\beta\sqrt{\pi}}\int_0^\infty e^{-x^2/\beta^2}\cos\!\left(\frac{2\sqrt{p}}{\beta}\sinh x\right)\mathrm{d}x. \tag{A2}$$

With $\cos(a\sinh x) = \frac{2}{\pi}\int_0^\infty K_{i\tau}(a)\cosh(\frac{\pi\tau}{2})\cos(\tau x)\,\mathrm{d}\tau$ [43] and $y = 2\sqrt{p}/\beta$,

$$\phi = \frac{2}{\pi}\int_0^\infty e^{-\beta^2\tau^2/4}\cosh\!\left(\frac{\pi\tau}{2}\right)K_{i\tau}(y)\,\mathrm{d}\tau, \tag{A3}$$

with $\phi(y=0) = 1$ and $\phi(y\to\infty) = 0$.

Mean eddy viscosity. From $\ln Z = \int_0^\infty (e^{-p} - e^{-pZ})\mathrm{d}p/p$, $\langle \ln Z\rangle = \int_0^\infty (e^{-p} - \phi)\mathrm{d}p/p$. Differentiating at fixed p and changing to y,

$$\langle \nu_e \rangle = -2\int_0^\infty \frac{\beta\,\partial_\beta \phi|_y}{y}\mathrm{d}y - 2, \tag{A4}$$

where $-2 = 2[\phi(\infty) - \phi(0)]$. With Eq. (A3) and $\int_0^\infty K_{i\tau}(y)\,y^{-1}\mathrm{d}y = \pi/[2\tau\sinh(\pi\tau/2)]$,

$$\langle \nu_e \rangle = \int_0^\infty \beta^2\tau\, e^{-\beta^2\tau^2/4}\coth\!\left(\frac{\pi\tau}{2}\right)\mathrm{d}\tau - 2, \tag{A5}$$

and $\tau = 2x/\beta$ gives Eq. (23). Equation (A1) for $n = 1$–3, $\langle \ln Z\rangle = \int_0^s \langle \nu_e\rangle\,\mathrm{d}s'/s'$ and Eq. (23) over $0.3 \leqslant s \leqslant 100$ agree with Monte Carlo simulation.

APPENDIX B: LONG-WAVE LIMIT OF THE SPECTRAL EDDY VISCOSITY FOR A GENERAL FILTER

Write $1/[p(k-p)] = k^{-1}[1/p + 1/(k-p)]$; by the symmetry $p \leftrightarrow k-p$ the two terms contribute equally to Eq. (39), so the principal-value integral is $2k^{-1}\mathrm{P}\int \hat{G}(p)\hat{G}(k-p)\,p^{-1}\mathrm{d}p$. With $\hat{G}(k-p) = \hat{G}(p) - k\hat{G}'(p) + O(k^2)$ ($\hat{G}$ even) and $\mathrm{P}\int \hat{G}^2 p^{-1}\mathrm{d}p = 0$, the integral is $-2\int_{-\infty}^{\infty}\hat{G}\hat{G}'p^{-1}\mathrm{d}p + O(k) = -2\int_0^\infty p^{-1}\mathrm{d}\hat{G}^2 + O(k)$. Inserting into Eq. (39) with $\hat{G}(0) = 1$ gives Eq. (40). For the sharp cutoff $\mathrm{d}\hat{G}^2/\mathrm{d}p = -\delta(p-K)$ and the result is $u_0/(\pi K)$. Numerically, principal values are removed by symmetrizing about each pole, $\mathrm{P}\int f(p)p^{-1}\mathrm{d}p = \int_0^\infty [f(t) - f(-t)]t^{-1}\mathrm{d}t$.

References

[1] A. N. Kolmogorov, Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences **434**, 9 (1991).
[2] A. N. Kolmogorov, Proceedings of the Royal Society of London. Series A: Mathematical and Physical Sciences **434**, 15 (1991).
[3] U. Frisch, *Turbulence* (Cambridge University Press, 1995).
[4] S. B. Pope, *Turbulent Flows* (Cambridge University Press, 2000).
[5] K. R. Sreenivasan, Physics of Fluids **7**, 2778 (1995).
[6] P. K. Yeung and Y. Zhou, Physical Review E **56**, 1746 (1997).
[7] T. Ishihara, T. Gotoh, and Y. Kaneda, Annual Review of Fluid Mechanics **41**, 165 (2009).

[8] T. de Kármán and L. Howarth, Proceedings of the Royal Society of London. Series A - Mathematical and Physical Sciences **164**, 192 (1938).
[9] W. Heisenberg, Zeitschrift für Physik **124**, 628 (1948).
[10] L. S. G. Kovasznay, Journal of the Aeronautical Sciences **15**, 745 (1948).
[11] Y.-H. Pao, The Physics of Fluids **8**, 1063 (1965).
[12] R. H. Kraichnan, Journal of Fluid Mechanics **5**, 497 (1959).
[13] S. A. Orszag, Journal of Fluid Mechanics **41**, 363 (1970).
[14] V. Yakhot and S. A. Orszag, Journal of Scientific Computing **1**, 3 (1986).
[15] L. Canet, B. Delamotte, and N. Wschebor, Physical Review E **93**, 063101 (2016).
[16] Z. Xu, Scientific Reports (2026), 10.1038/s41598-026-69825-4, preprint: Research Square, doi:10.21203/rs.3.rs-9674983/v1.
[17] R. H. Kraichnan, The Physics of Fluids **8**, 575 (1965).
[18] Y. Kaneda, The Physics of Fluids **29**, 701 (1986).
[19] A. Alexakis and L. Biferale, Physics Reports **767-769**, 1 (2018).
[20] J. Smagorinsky, Monthly Weather Review **91**, 99 (1963).
[21] C. Meneveau and J. Katz, Annual Review of Fluid Mechanics **32**, 1 (2000).
[22] R. H. Kraichnan, Journal of the Atmospheric Sciences **33**, 1521 (1976).
[23] J.-P. Chollet and M. Lesieur, Journal of the Atmospheric Sciences **38**, 2747 (1981).
[24] J. A. Domaradzki, R. W. Metcalfe, R. S. Rogallo, and J. J. Riley, Physical Review Letters **58**, 547 (1987).
[25] S. Cerutti, C. Meneveau, and O. M. Knio, Journal of Fluid Mechanics **421**, 307 (2000).
[26] A. Bensoussan, J.-L. Lions, and G. Papanicolaou, *Asymptotic Analysis for Periodic Structures* (American Mathematical Society, 2011).
[27] D. W. McLaughlin, G. C. Papanicolaou, and O. R. Pironneau, SIAM Journal on Applied Mathematics **45**, 780 (1985).
[28] M. Avellaneda and A. J. Majda, Communications in Mathematical Physics **138**, 339 (1991).
[29] A. Fannjiang and G. Papanicolaou, SIAM Journal on Applied Mathematics **54**, 333 (1994).
[30] A. J. Majda and P. R. Kramer, Physics Reports **314**, 237 (1999).
[31] G. A. Pavliotis and A. M. Stuart, *Multiscale Methods: Averaging and Homogenization* (Springer New York, 2008).
[32] B. Dubrulle and U. Frisch, Physical Review A **43**, 5355 (1991).
[33] A. Wirth, S. Gama, and U. Frisch, Journal of Fluid Mechanics **288**, 249 (1995).
[34] E. Perlman, R. Burns, Y. Li, and C. Meneveau, in *Proceedings of the 2007 ACM/IEEE Conference on Supercomputing* (2007) pp. 1–11.
[35] Y. Li, E. Perlman, M. Wan, Y. Yang, C. Meneveau, R. Burns, S. Chen, A. Szalay, and G. Eyink, Journal of Turbulence **9**, N31 (2008).
[36] A. A. Oberai, C. E. Colosqui, and J. Wanderer, International Journal for Multiscale Computational Engineering **6**, 299 (2008).
[37] U. Frisch, P.-L. Sulem, and M. Nelkin, Journal of Fluid Mechanics **87**, 719 (1978).
[38] J. M. Burgers, in *Advances in Applied Mechanics* (Elsevier, 1948) pp. 171–199.
[39] J. Bec and K. Khanin, Physics Reports **447**, 1 (2007).
[40] G. Oshanin, A. Mogutov, and M. Moreau, Journal of Statistical Physics **73**, 379 (1993).
[41] A. Comtet, C. Monthus, and M. Yor, Journal of Applied Probability **35**, 255 (1998).
[42] H. Matsumoto and M. Yor, Probability Surveys **2**, 312 (2005).
[43] S. B. Yakubovich, *Index Transforms* (World Scientific, 1996).
[44] L. Beghin and E. Orsingher, Lithuanian Mathematical Journal **39**, 157 (1999).
[45] A. Leonard, in *Advances in Geophysics* (Elsevier, 1975) pp. 237–248.
[46] R. A. Clark, J. H. Ferziger, and W. C. Reynolds, Journal of Fluid Mechanics **91**, 1 (1979).
[47] J. A. Domaradzki and R. S. Rogallo, Physics of Fluids A: Fluid Dynamics **2**, 413 (1990).
[48] D. C. Leslie and G. L. Quarini, Journal of Fluid Mechanics **91**, 65 (1979).
[49] M. D. Love, Journal of Fluid Mechanics **100**, 87 (1980).
[50] R. Maulik and O. San, Journal of Computational and Applied Mathematics **327**, 12 (2018).
[51] S. Kida, Journal of Fluid Mechanics **93**, 337 (1979).
[52] L. Prandtl, ZAMM - Journal of Applied Mathematics and Mechanics / Zeitschrift für Angewandte Mathematik und Mechanik **5**, 136 (1925).
[53] G. I. Taylor, Proceedings of the London Mathematical Society **s2-20**, 196 (1922).
[54] G. K. Batchelor, Mathematical Proceedings of the Cambridge Philosophical Society **47**, 359 (1951).
[55] R. H. Kraichnan, Journal of Fluid Mechanics **47**, 525 (1971).
[56] V. Borue and S. A. Orszag, Journal of Fluid Mechanics **366**, 1 (1998).
[57] S. Liu, C. Meneveau, and J. Katz, Journal of Fluid Mechanics **275**, 83 (1994).
[58] Z. Xu, "Eddy viscosity from stochastic homogenization: manuscript and figure data," Zenodo, doi:10.5281/zenodo.24096071 (2026).